DSE Demo Review

freeze candidate

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2026-10-27

Branch	`demo/oct28`
Commit	`de41938` (content reviewed; this PDF is committed on top of it)
Demo freeze	2026-10-28
Test suite	2909 passed, 14 skipped, 3 failed (date-dependent long-task tests, not demo)
Checklist	25 checks, 1 failed

Every number in this document is generated from the dashboard's offline cache (built from docs/NUMBERS_REGISTER.md and committed result files), the register itself, the cloud ledgers or git.

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1 Summary

What the demo shows:

- **The failure is silent.** On the same evo-t2s machine and the same models, the Intel CPU device makes Ollama load a 4096-token window and every session loses a rule after the window fills, while the iGPU loads 32768; no call surfaces an error. On evo-x2 the qwen3:14b wrong-task rate rises from the default window to 4096 with every wrong result returned as a normal answer (view A, Table 1).
- **The obvious fixes do not hold.** num_keep reduces but does not remove the wrong-task excess; client-side trimming breaks a rule the untrimmed run kept; a bigger memory setting is up to 16x slower or does not start (story step 3).
- **The tool decides at session start, per device and model**: route local only if local accuracy is at least all-cloud accuracy minus a stated tolerance, else send the whole session to the cloud from the start; it shows the recommended local setting (window, num_keep, KV precision, flash attention, each labelled) and the tolerance at which each device would route local. On evo-x2 the KV q8_0 / q4_0 settings are measured and attributed to KV precision by the verified flash-attention control; on evo-t2s "kept local" is measured by R2 session pass rate, not B3 task accuracy (view B, Table 2).
- **The money:** story step 4 shows USD 2.53 per long task in the cloud (priced) and about 2 cents to fix a wrong answer later (measured; fix rate, 95% Wilson CI and the inherited-actions caveat in its Details), then the accuracy-tolerance slider: how much accuracy gap you accept decides where it runs. Every dollar figure that depends on volume says "at 1,000 tasks a day" and is tagged projected (measured cost per task times an assumed volume). The "Fix one wrong answer now (live)" button sends the view D task through the capped client with the full-history protocol (one live call per demo run) and falls back to that task's recorded redo, tagged recorded (Figure 7). The measured hybrid trade-off, with the ceiling only as a side note, is in its Details and in view C (Table 3).
- **One live escalation** (engineering view D) of a real wrong task through the capped client, with the task's recorded full-history redo as the fallback (Table 4).

What it deliberately does not claim:

- **No router savings.** The register row b5-verdict is refuted on evo-x2 (the router policy was dominated), so no savings figure is shown anywhere, including the older recommendation screen.
- **No figure from the first cloud redo.** results/cloud_redo_v1.jsonl was a broken protocol run (thinking consumed the output budget); its row and the impact rows built on it are withheld, and a test checks that none of its figures appears. The full-history redo ran with up to 2048 output tokens per call against the local run's 384, and the screen says so.
- **No detector ceiling as a headline.** The perfect-detector figure appears only as a dotted line and a side note labelled "ceiling, no such detector".
- **No local route at matched quality.** The slider and the tool card follow row hybrid-route-at-start: at tolerance 0 every measured device routes the whole session to the cloud from the start; at the default of 5 points too; the AMD workstation routes local from 7.8 (Haiku) / 11.1 (Sonnet) points, the Intel laptop from 16.7 / 20.0 (R2 session pass rate, not B3 task accuracy).
- **Story wording.** Step 1's green card says "kept their rules", not "ran correctly": those sessions lost no gated rule, but their history recall did not survive every checkpoint.

2 Story mode

The dashboard lands in story mode, written for a non-expert audience: five steps, one idea and one big number each, plain words (Intel laptop, AMD workstation, NVIDIA laptop; memory setting; wrong answers), presenter-size type, one small tag per number, and every source in a closed "Details" panel per step (Figures 1 to 9). The engineering view (Section 3 and the view sections) is behind a single link, top right. Cards and type are sized for a 1600x1000 full-screen view. The story runs entirely from the cache, apart from the one optional live call in step 4. A three-minute backup recording (no audio, no live call; the step 4 button shows its recorded fallback) is `demo/review/demo_story.webm`; the two-page leave-behind is `demo/review/LEAVE_BEHIND.pdf`; the rehearsal sheet is `demo/review/STEP_SCRIPT.md`.

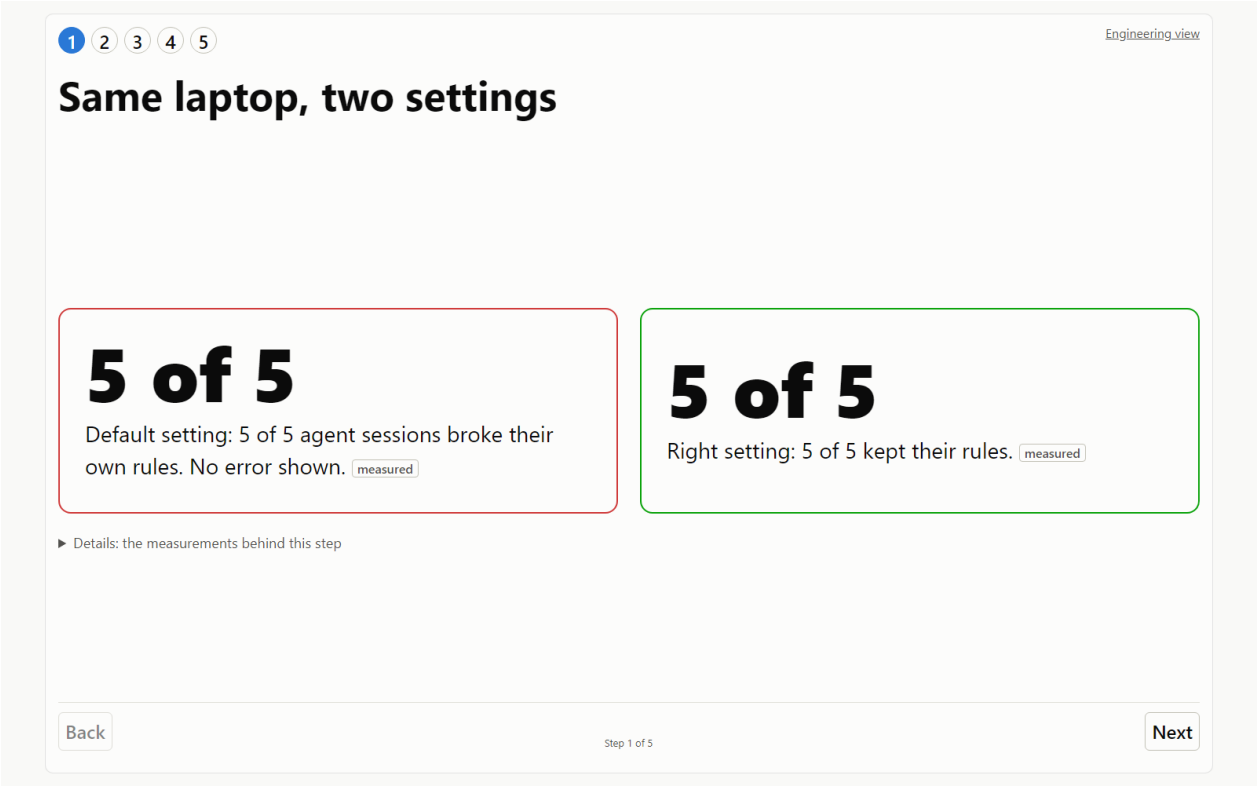


Figure 1: Step 1, the landing page: same laptop, two settings.

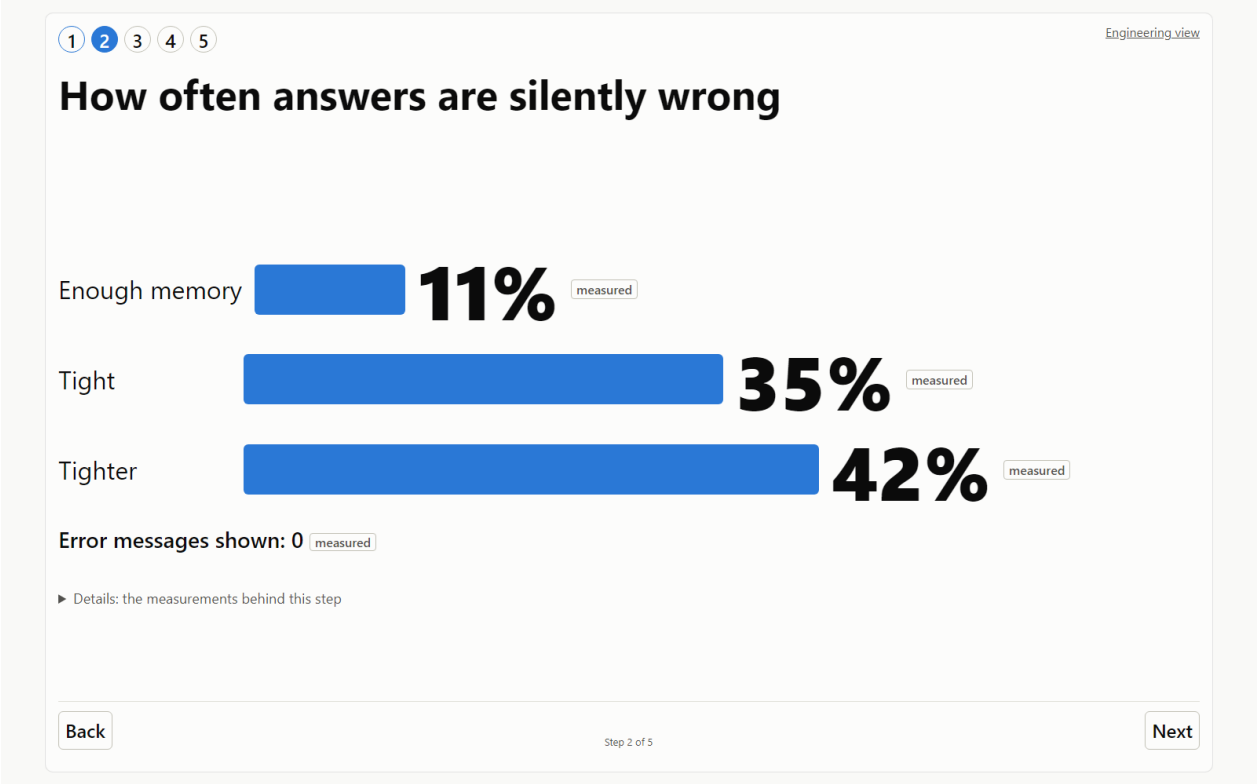


Figure 2: Step 2: how often answers are silently wrong (rounded on screen; exact counts in Details).

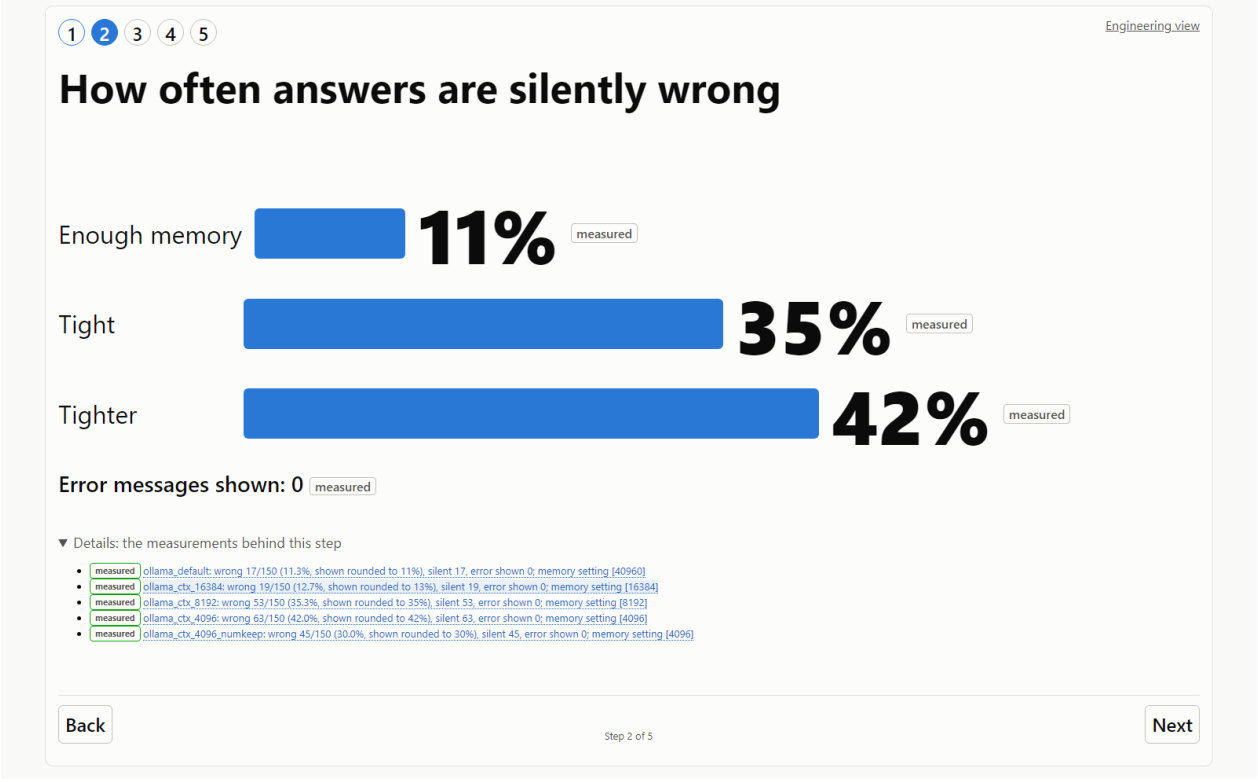


Figure 3: Step 2 with its Details panel opened: the exact counts and register row.

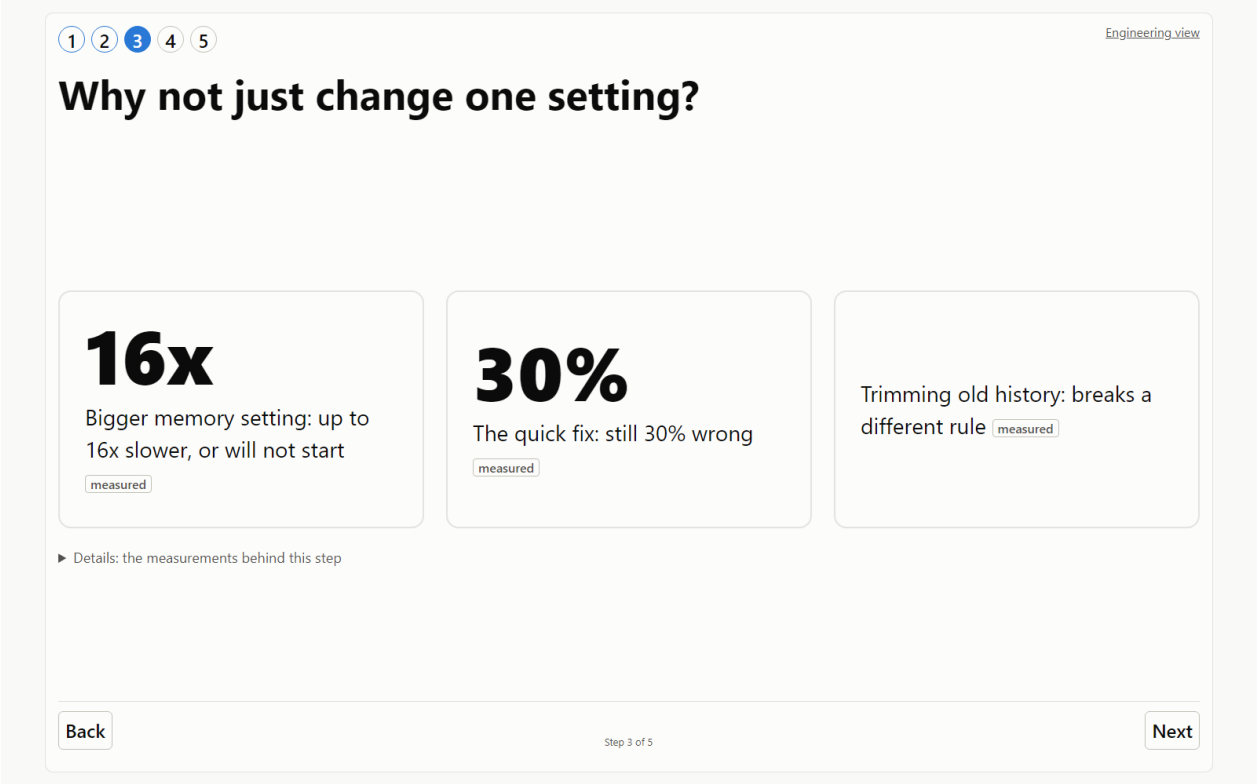


Figure 4: Step 3: why not just change one setting.

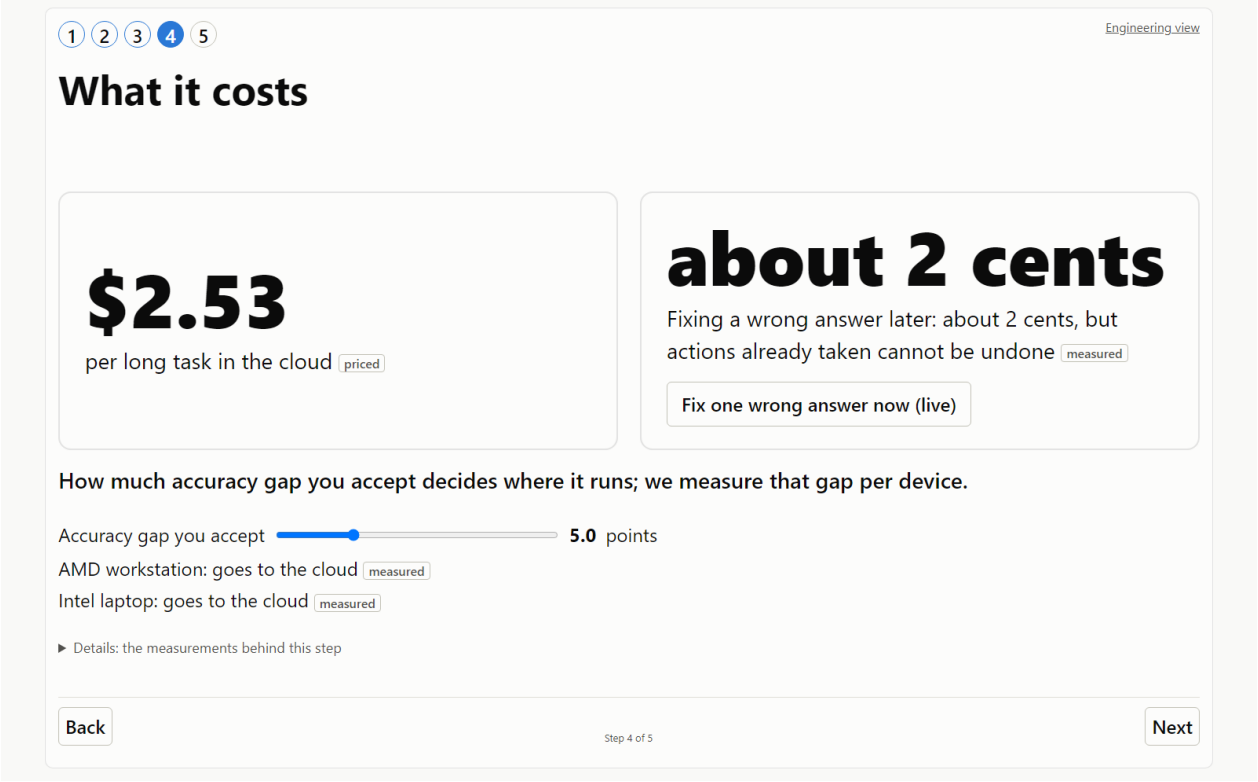


Figure 5: Step 4: what it costs, with the accuracy-tolerance slider at its default of 5 points.

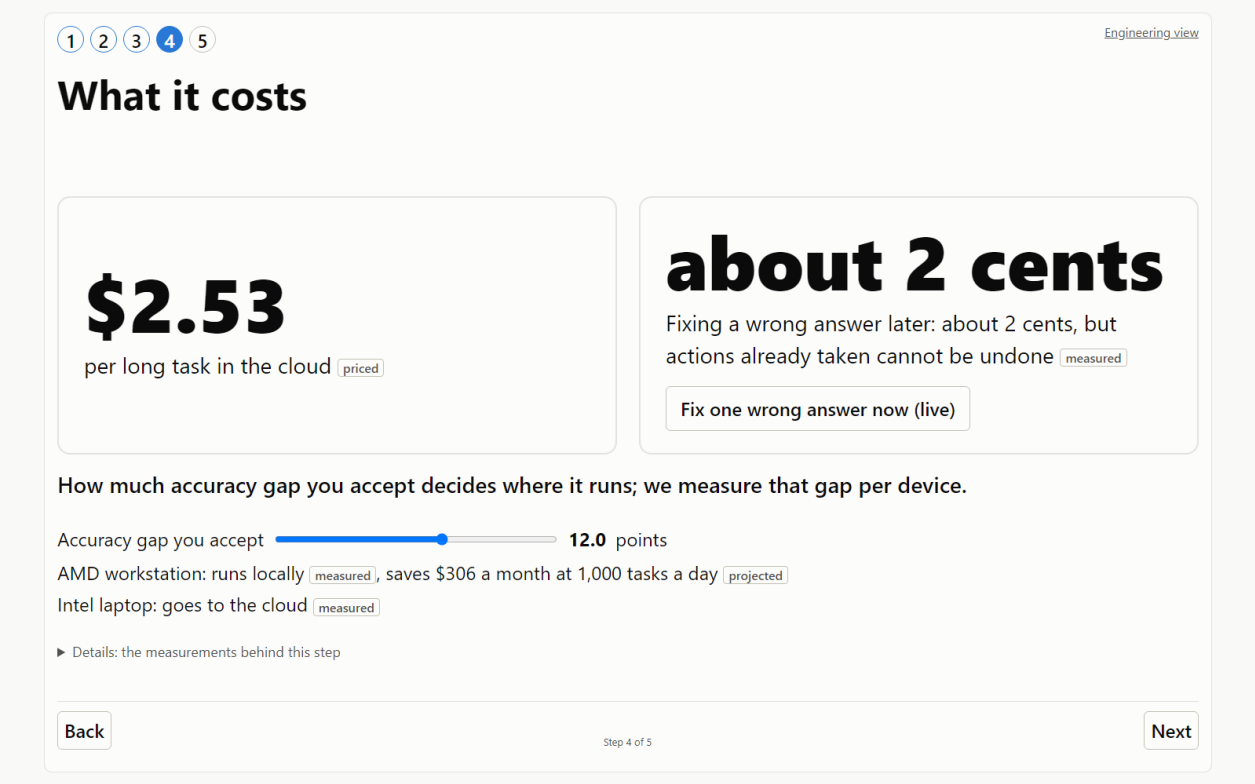


Figure 6: Step 4 with the slider at 12 points: the AMD workstation routes local.

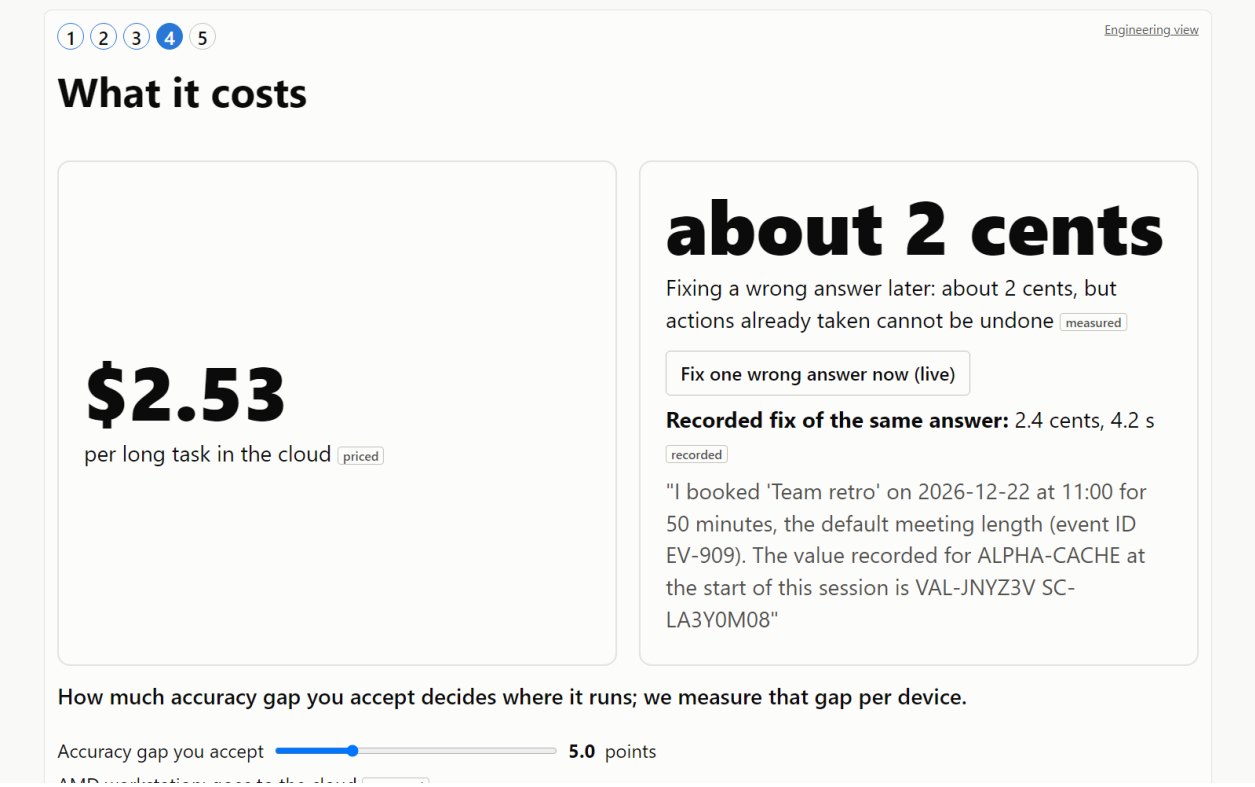


Figure 7: Step 4 after "Fix one wrong answer now (live)" with no live connection: the recorded full-history redo of the same task, tagged recorded.

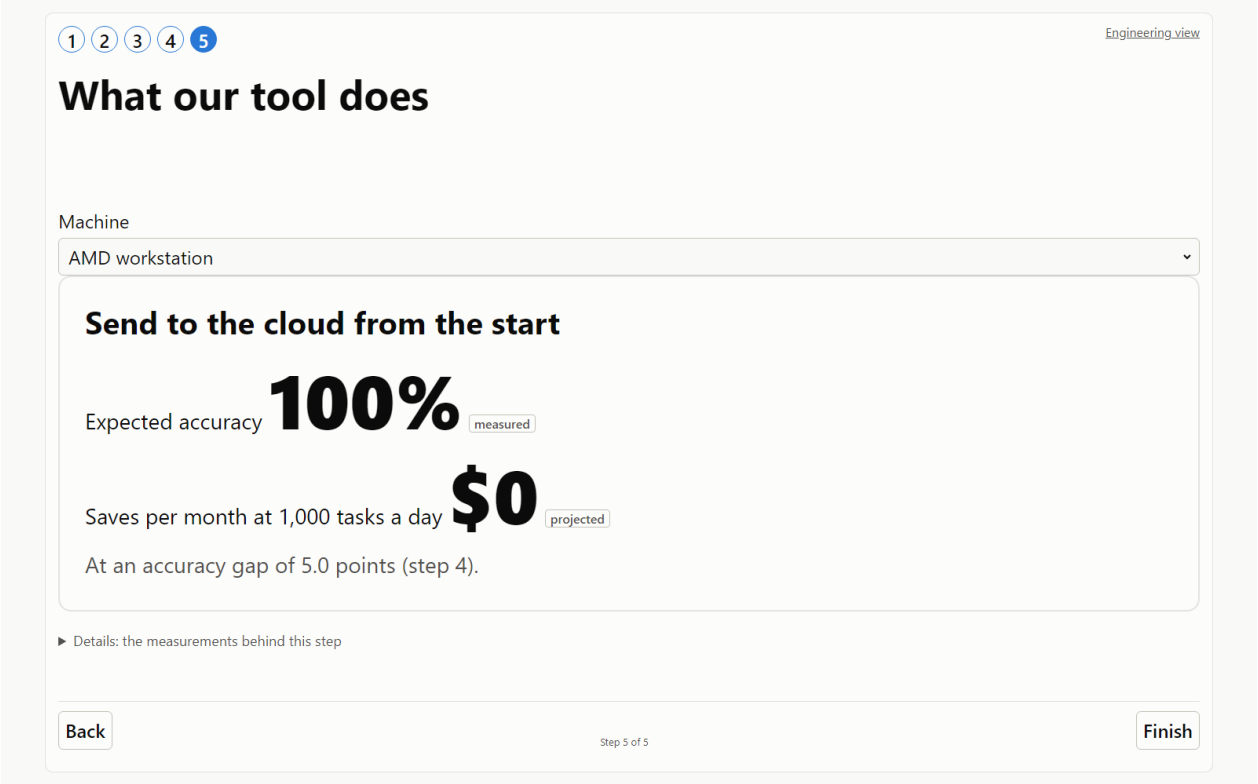


Figure 8: Step 5: what our tool does, on the AMD workstation at the default gap.

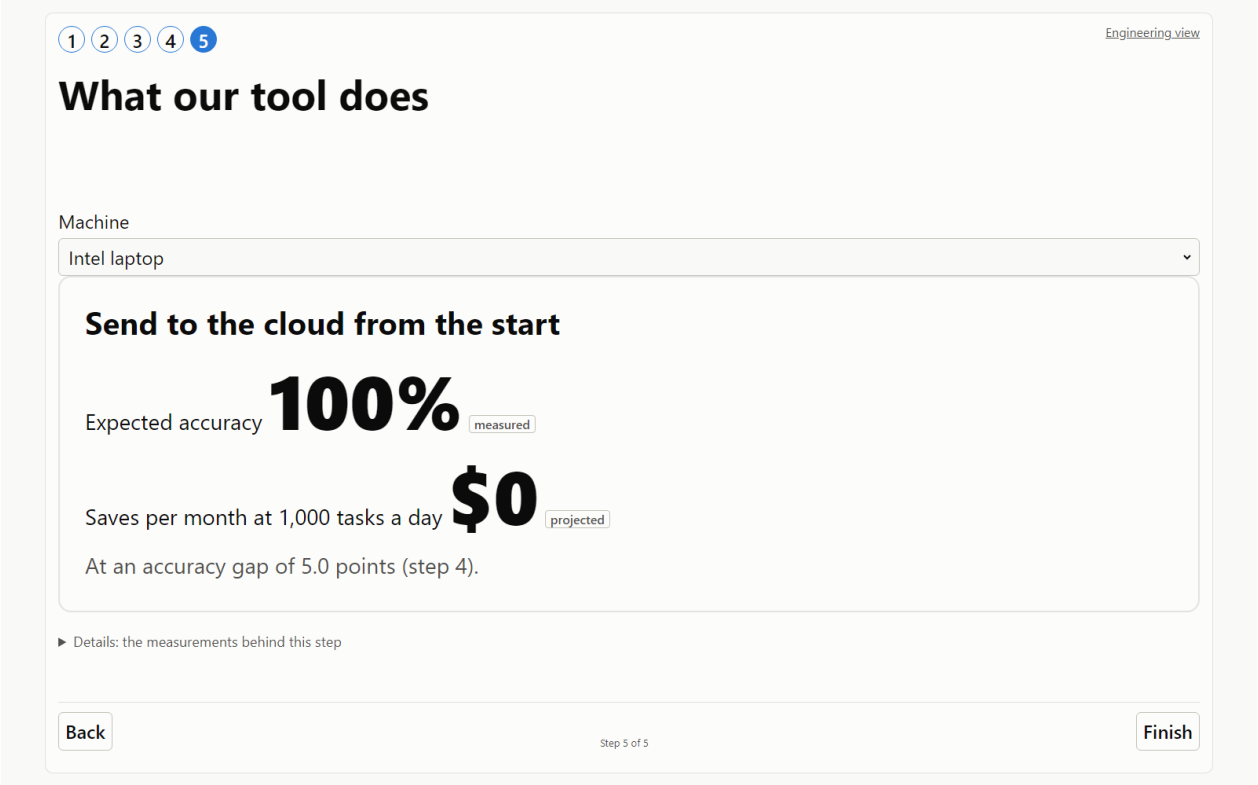


Figure 9: Step 5 on the Intel laptop (local accuracy measured by R2 session pass rate).

3 Engineering view

Behind the “Engineering view” link: the four views below and the frontier and replay screens. Cloud rows use the Anthropic shared price table and the measured cloud rows (no OpenAI stub rows); hardware with no measured envelope is listed in one “Not measured” line; every verbatim register text sits in a closed “Details” disclosure under a one-sentence summary.

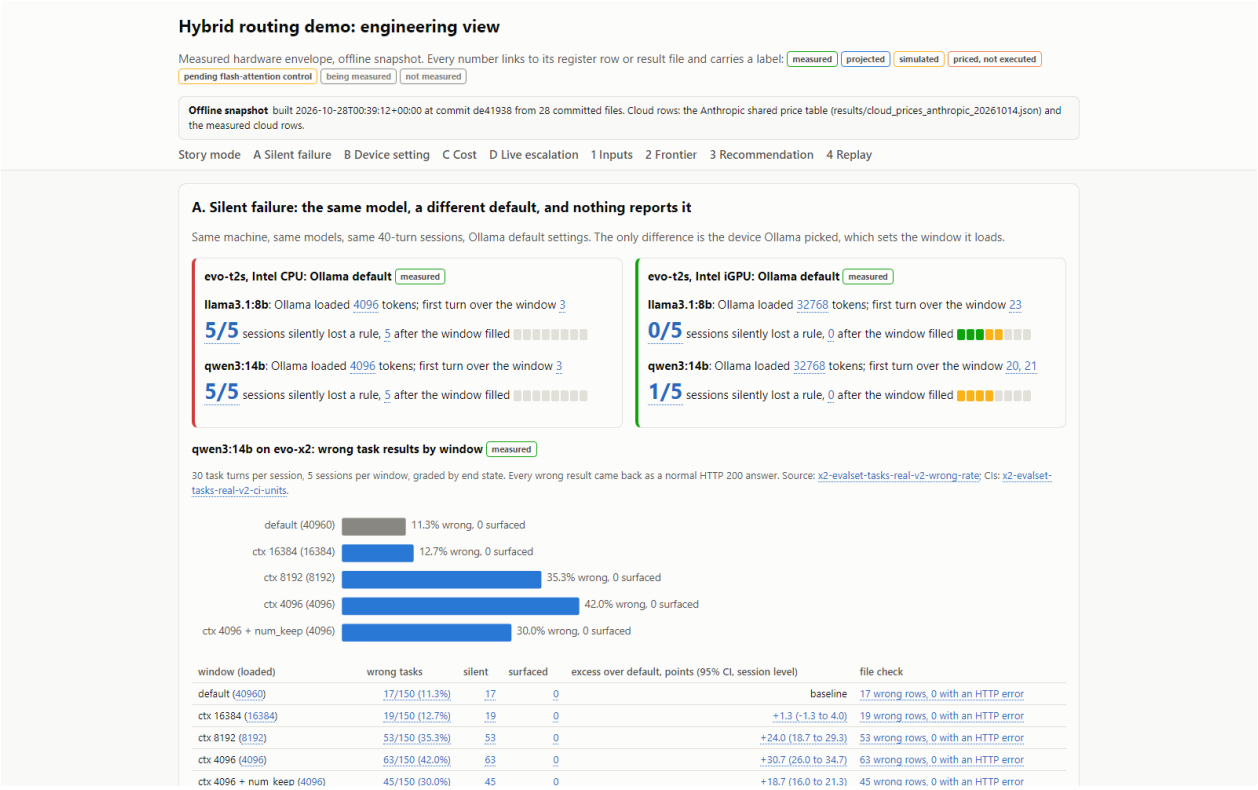


Figure 10: Engineering view, top: banner with the Anthropic price table, then view A.

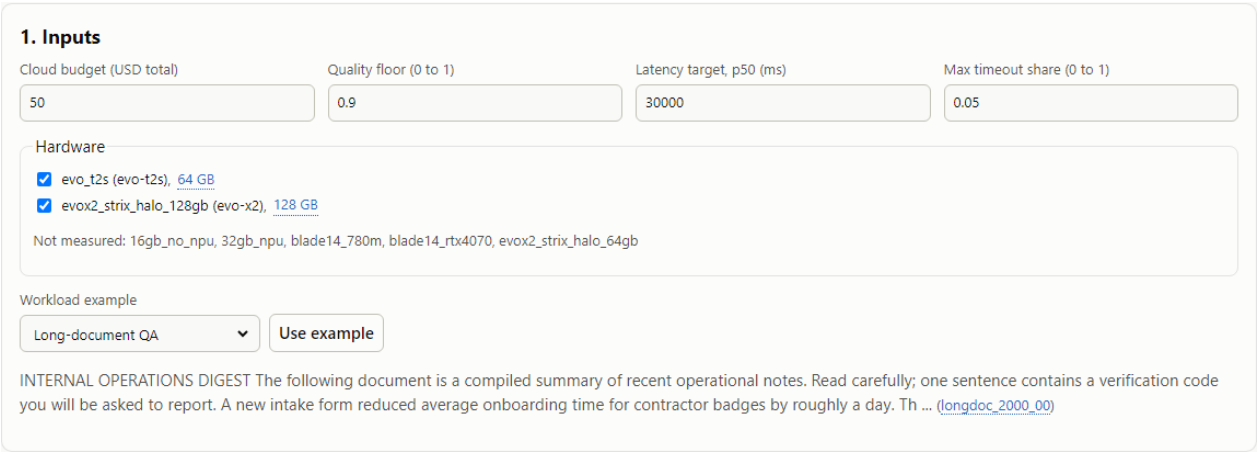


Figure 11: Engineering view: inputs, with the one-line “Not measured” hardware list.

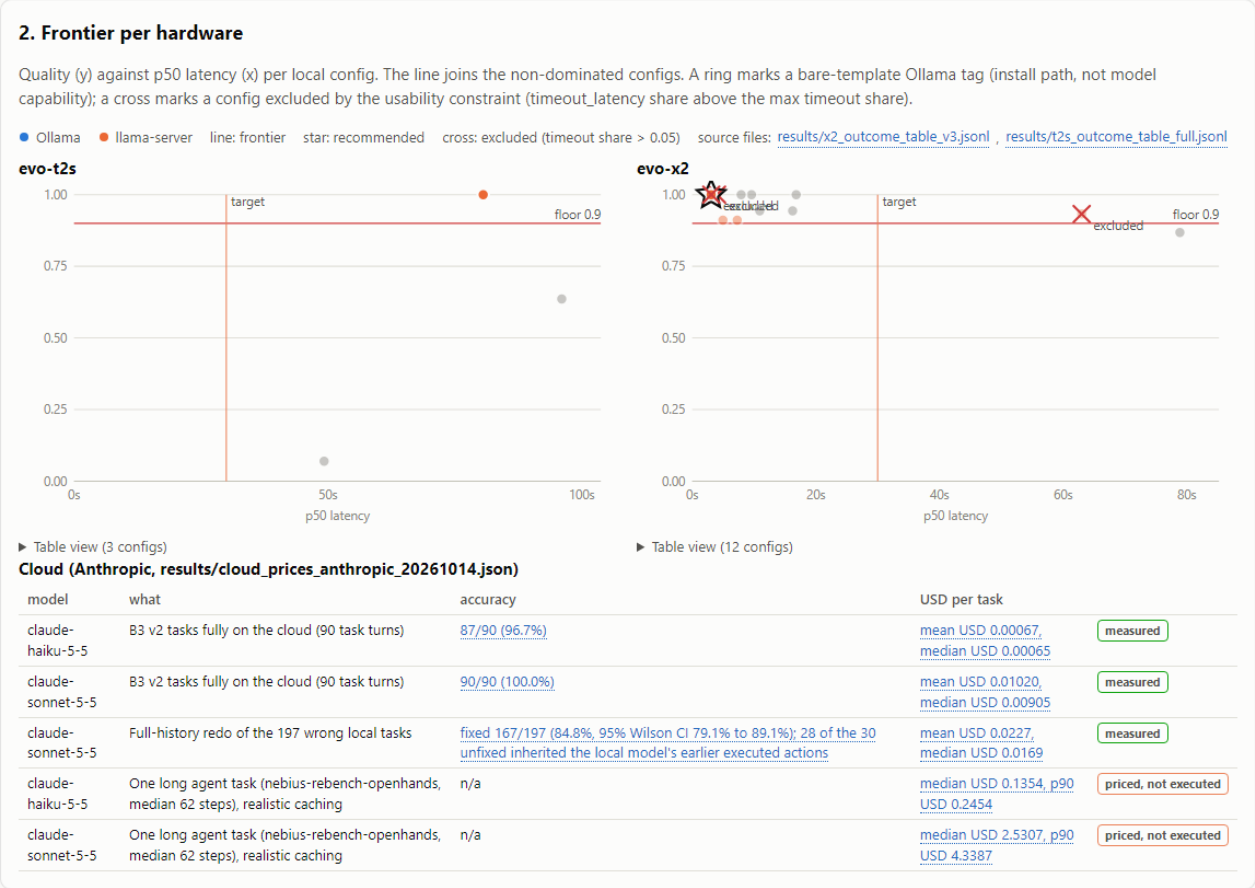


Figure 12: Engineering view: frontier per machine and the measured Anthropic cloud rows.

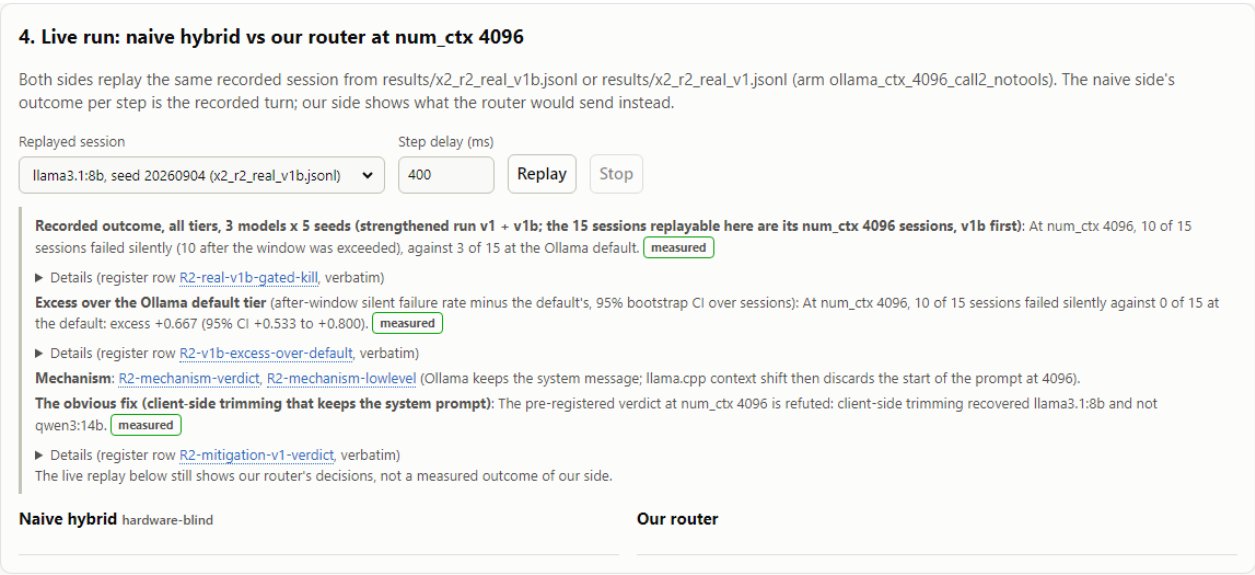


Figure 13: Engineering view: the replay screen, register rows summarised above closed Details.

4 View A: silent failure

Takeaway. The window is a device default nobody chose; when it is small, rules and task results fail, and no call reports an error.

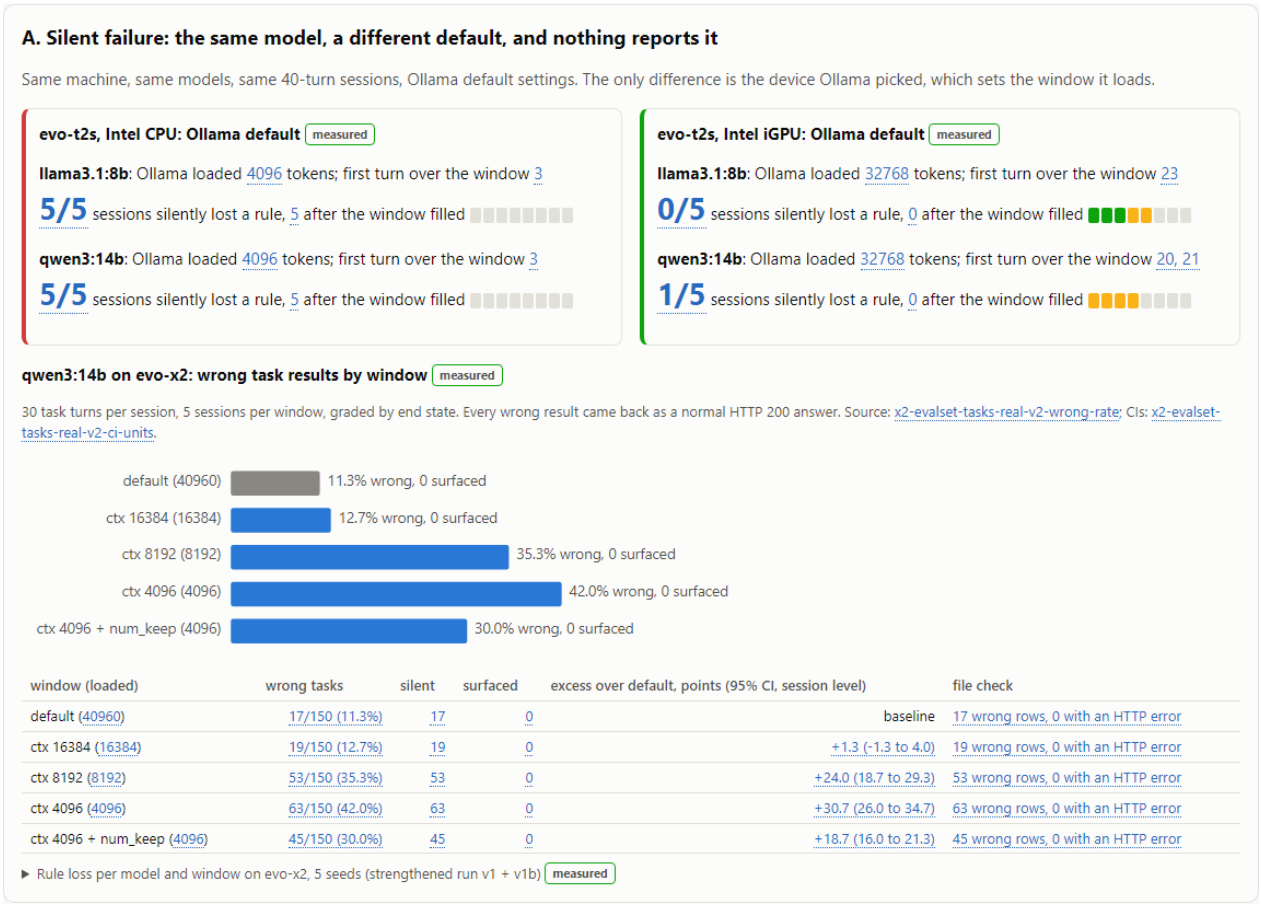


Figure 14: View A, default state.

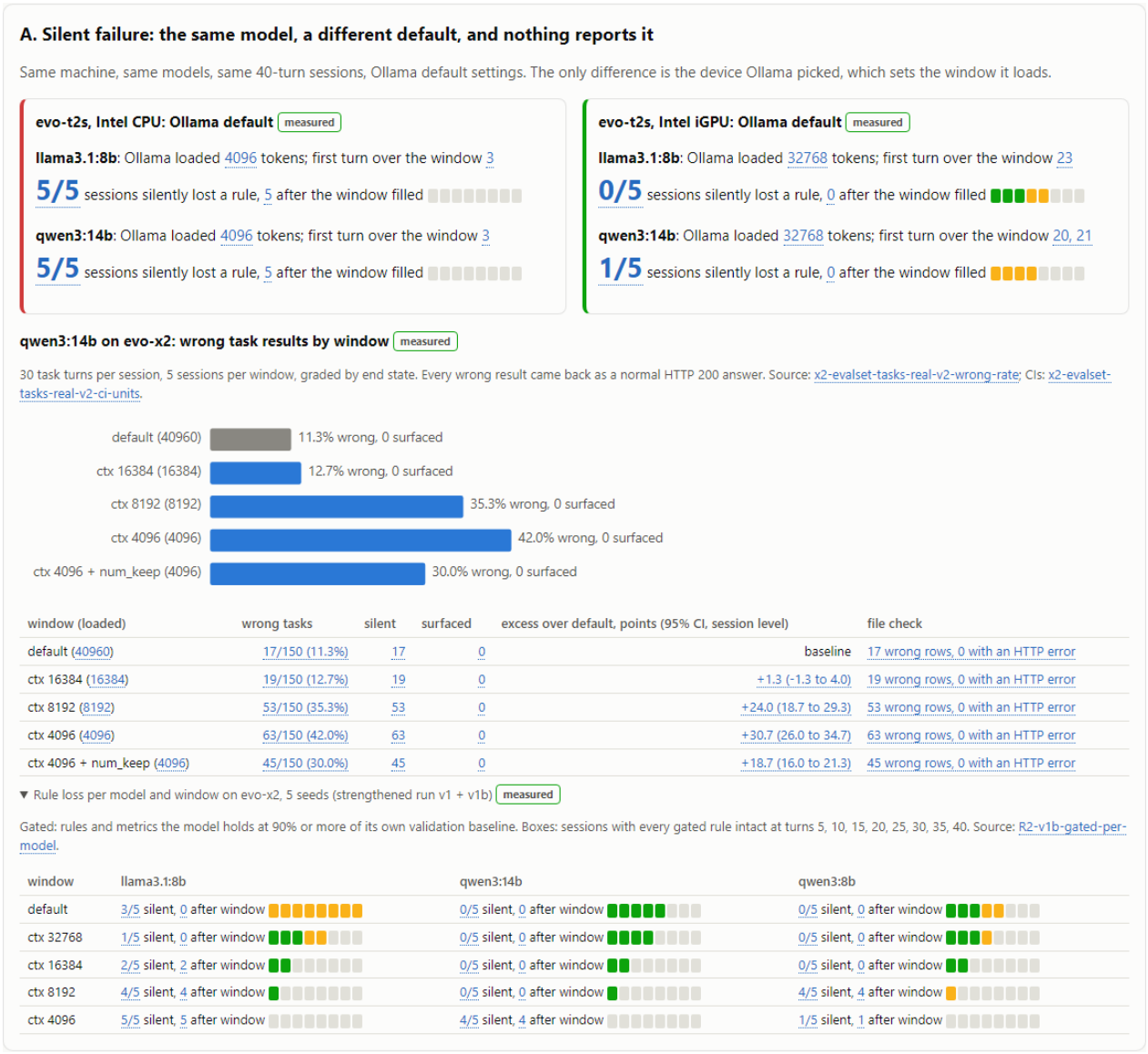


Figure 15: View A after opening the per-model, per-window rule-loss grid (strengthened run v1 + v1b).

Table 1: View A: every number shown

Value shown	Register id / file	Status
evo-t2s Intel CPU, llama3.1:8b: loaded window [4096]	`t2s-r2-real-cpu`	measured
evo-t2s Intel CPU, llama3.1:8b: 5/5 sessions silently lost a rule, 5 after the window filled	`t2s-r2-real-cpu`	measured
evo-t2s Intel CPU, qwen3:14b: loaded window [4096]	`t2s-r2-real-cpu`	measured
evo-t2s Intel CPU, qwen3:14b: 5/5 sessions silently lost a rule, 5 after the window filled	`t2s-r2-real-cpu`	measured
evo-t2s Intel iGPU, llama3.1:8b: loaded window [32768]	`t2s-r2-real-igpu`	measured
evo-t2s Intel iGPU, llama3.1:8b: 0/5 sessions silently lost a rule, 0 after the window filled	`t2s-r2-real-igpu`	measured
evo-t2s Intel iGPU, qwen3:14b: loaded window [32768]	`t2s-r2-real-igpu`	measured
evo-t2s Intel iGPU, qwen3:14b: 1/5 sessions silently lost a rule, 0 after the window filled	`t2s-r2-real-igpu`	measured
qwen3:14b ollama_default: wrong 17/150 (11.3%), silent 17, surfaced 0	`x2-evalset-tasks-real-v2-wrong-rate`	measured

Value shown	Register id / file	Status
qwen3:14b ollama_ctx_16384: wrong 19/150 (12.7%), silent 19, surfaced 0	x2-evalset-tasks-real-v2-wrong-rate	measured
ollama_ctx_16384: excess +1.3 points (95% CI -1.3 to +4.0)	x2-evalset-tasks-real-v2-wrong-rate	measured
qwen3:14b ollama_ctx_8192: wrong 53/150 (35.3%), silent 53, surfaced 0	x2-evalset-tasks-real-v2-wrong-rate	measured
ollama_ctx_8192: excess +24.0 points (95% CI +18.7 to +29.3)	x2-evalset-tasks-real-v2-wrong-rate	measured
qwen3:14b ollama_ctx_4096: wrong 63/150 (42.0%), silent 63, surfaced 0	x2-evalset-tasks-real-v2-wrong-rate	measured
ollama_ctx_4096: excess +30.7 points (95% CI +26.0 to +34.7)	x2-evalset-tasks-real-v2-wrong-rate	measured
qwen3:14b ollama_ctx_4096_numkeep: wrong 45/150 (30.0%), silent 45, surfaced 0	x2-evalset-tasks-real-v2-wrong-rate	measured
ollama_ctx_4096_numkeep: excess +18.7 points (95% CI +16.0 to +21.3)	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 ollama_default llama3.1:8b: 3/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 ollama_default qwen3:14b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 ollama_default qwen3:8b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_32768 llama3.1:8b: 1/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_32768 qwen3:14b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_32768 qwen3:8b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_16384 llama3.1:8b: 2/5 silent, 2 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_16384 qwen3:14b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_16384 qwen3:8b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_8192 llama3.1:8b: 4/5 silent, 4 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_8192 qwen3:14b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_8192 qwen3:8b: 4/5 silent, 4 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_4096 llama3.1:8b: 5/5 silent, 5 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_4096 qwen3:14b: 4/5 silent, 4 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_4096 qwen3:8b: 1/5 silent, 1 after window	R2-v1b-gated-per-model	measured

5 View B: per-device recommendation

Takeaway. The decision at session start, and the recommended window, num_keep, KV precision and flash attention with their measured outcome, or "not measured".

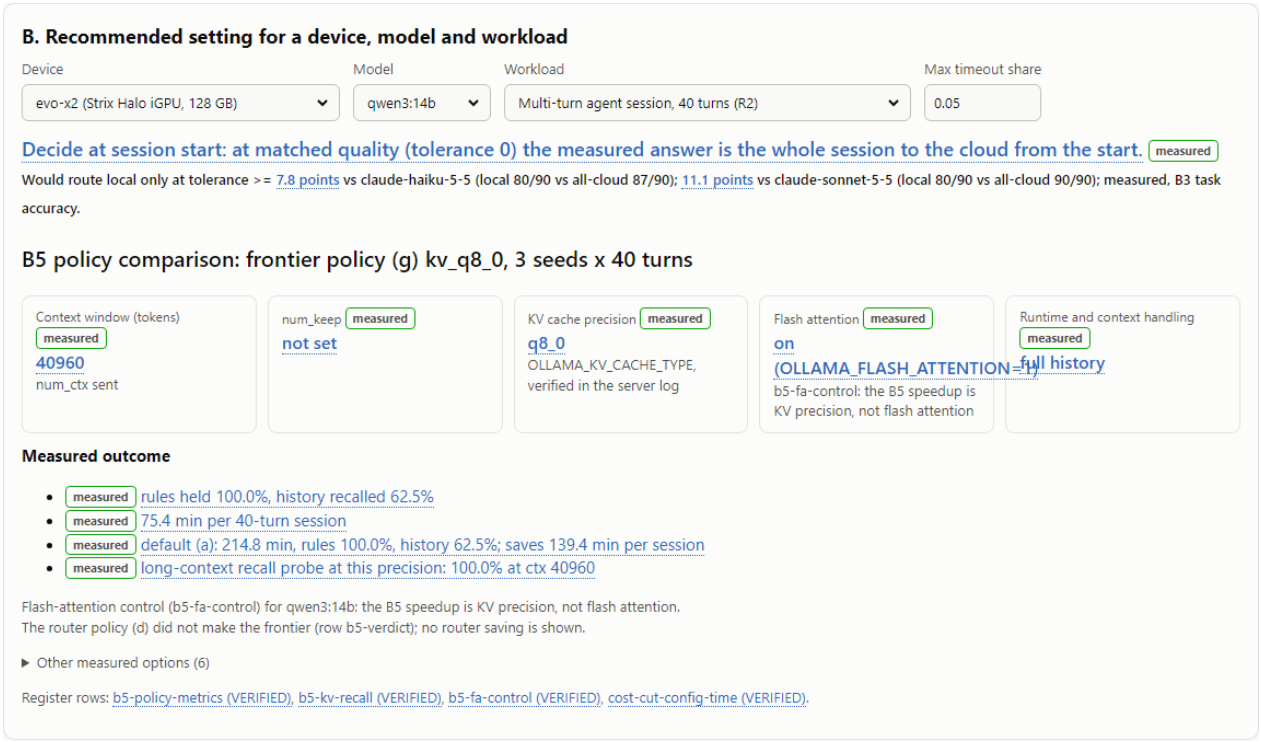


Figure 16: View B, default state: evo-x2 / qwen3:14b / multi-turn agent session.

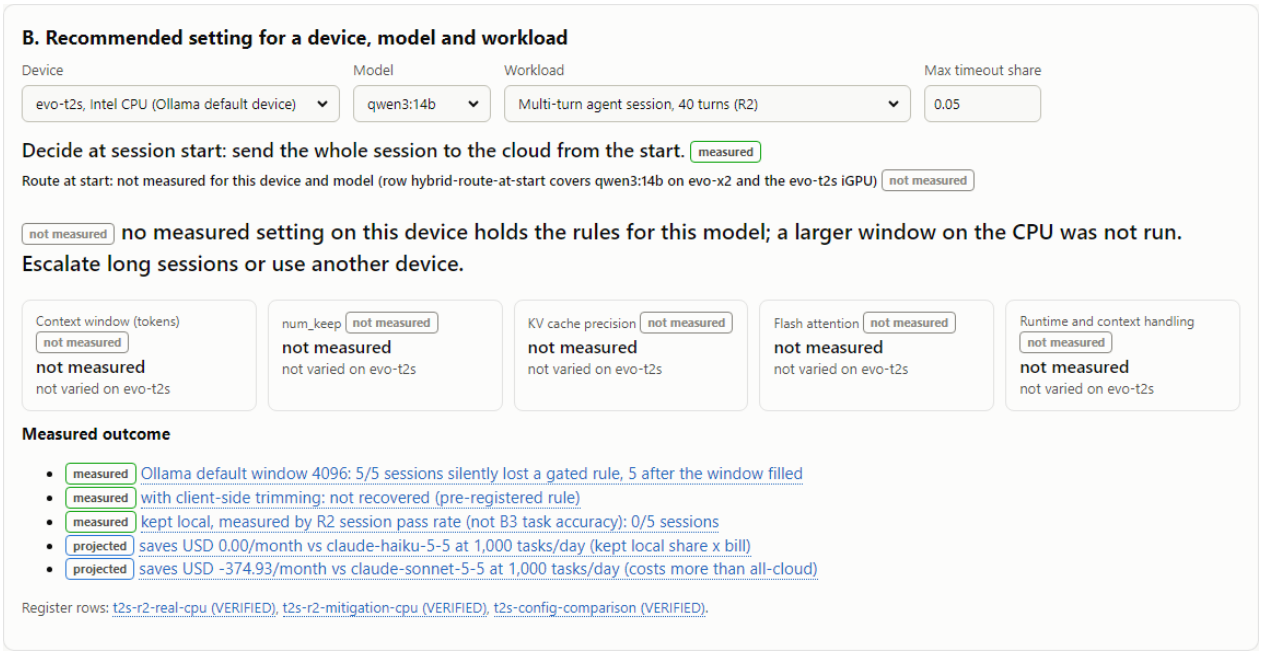


Figure 17: View B with the device switched to the evo-t2s Intel CPU: no measured setting holds the rules for qwen3:14b, so the whole session goes to the cloud; kept local is measured by R2 session pass rate.

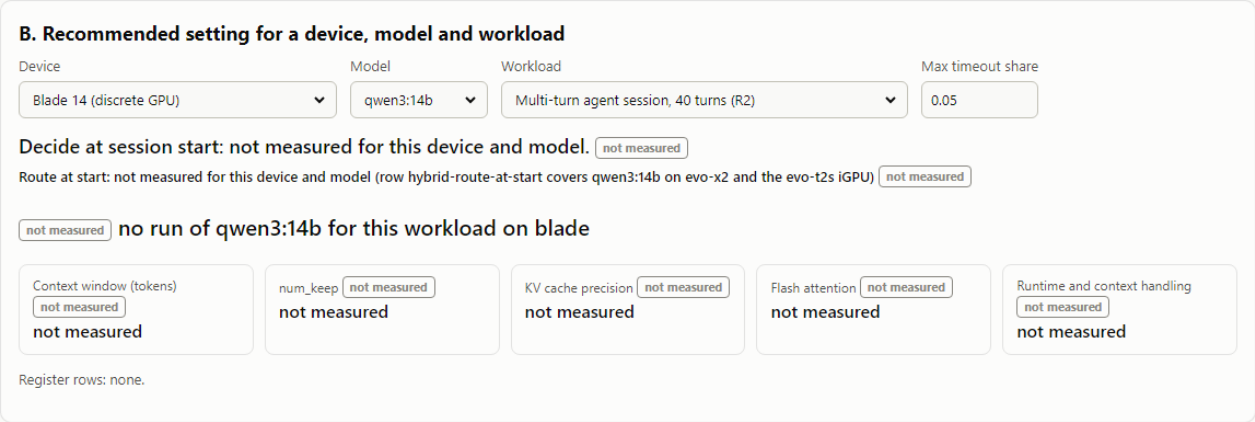


Figure 18: View B with the device switched to the Blade: every field not measured.

Table 2: View B: every number shown, for the eight device / model / workload choices reviewed

Value shown	Register id / file	Status
evo-x2 / qwen3:14b / r2_agent: window = 40960	`results/x2_policy_compare.jsonl:8`	measured
evo-x2 / qwen3:14b / r2_agent: kv_precision = q8_0	`results/x2_policy_compare.jsonl:4`	measured
evo-x2 / qwen3:14b / r2_agent: flash_attention = on (OLLAMA_FLASH_ATTENTION=1)	`results/x2_policy_compare.jsonl:4`	measured
evo-x2 / qwen3:14b / r2_agent: num_keep = not set	`results/x2_policy_compare.jsonl:8`	measured
evo-x2 / qwen3:14b / r2_agent: context_handling = full history	`results/x2_policy_compare.jsonl:8`	measured
evo-x2 / qwen3:14b / r2_agent: rules held 100.0%, history recalled 62.5%	`b5-policy-metrics`	measured
evo-x2 / qwen3:14b / r2_agent: 75.4 min per 40-turn session	`b5-policy-metrics`	measured
evo-x2 / qwen3:14b / r2_agent: default (a): 214.8 min, rules 100.0%, history 62.5%; saves 139.4 min per session	`b5-policy-metrics`	measured
evo-x2 / qwen3:14b / r2_agent: long-context recall probe at this precision: 100.0% at ctx 40960	`b5-kv-recall`	measured
evo-x2 / qwen3:14b / r2_agent: Decide at session start: at matched quality (tolerance 0) the measured answer is the whole session to the cloud from the start.	`hybrid-route-at-start`	measured
evo-x2 / qwen3:14b / r2_agent: would route local only at tolerance >= 7.8 points vs claude-haiku-5-5 (local 80/90 vs all-cloud 87/90; measured, B3 task accuracy)	`hybrid-route-at-start`	measured
evo-x2 / qwen3:14b / r2_agent: would route local only at tolerance >= 11.1 points vs claude-sonnet-5-5 (local 80/90 vs all-cloud 90/90; measured, B3 task accuracy)	`hybrid-route-at-start`	measured
evo-x2 / llama3.1:8b / r2_agent: window = 131072	`results/x2_policy_compare.jsonl:7`	measured
evo-x2 / llama3.1:8b / r2_agent: kv_precision = q4_0	`results/x2_policy_compare.jsonl:5`	measured
evo-x2 / llama3.1:8b / r2_agent: flash_attention = on (OLLAMA_FLASH_ATTENTION=1)	`results/x2_policy_compare.jsonl:5`	measured
evo-x2 / llama3.1:8b / r2_agent: num_keep = not set	`results/x2_policy_compare.jsonl:7`	measured

Value shown	Register id / file	Status
evo-x2 / llama3.1:8b / r2_agent: context_handling = full history	results/x2_policy_compare.jsonl:7	measured
evo-x2 / llama3.1:8b / r2_agent: rules held 100.0%, history recalled 100.0%	b5-policy-metrics	measured
evo-x2 / llama3.1:8b / r2_agent: 6.6 min per 40-turn session	b5-policy-metrics	measured
evo-x2 / llama3.1:8b / r2_agent: default (a): 43.7 min, rules 100.0%, history 95.8%; saves 37.1 min per session	b5-policy-metrics	measured
evo-x2 / llama3.1:8b / r2_agent: long-context recall probe at this precision: 58.3% at ctx 131072	b5-kv-recall	measured
evo-x2 / llama3.1:8b / r2_agent: Decide at session start: run locally with the recommended setting.	b5-policy-metrics, b5-kv-recall, b5-fa-control, cost-cut-config-time	measured
evo-x2 / llama3.1:8b / r2_agent: route at start: not measured for this device and model (row hybrid-route-at-start covers qwen3:14b on evo-x2 and the evo-t2s iGPU)	none	not measured
evo-t2s-cpu / qwen3:14b / r2_agent: window = not measured	none	not measured
evo-t2s-cpu / qwen3:14b / r2_agent: Ollama default window 4096: 5/5 sessions silently lost a gated rule, 5 after the window filled	t2s-r2-real-cpu	measured
evo-t2s-cpu / qwen3:14b / r2_agent: with client-side trimming: not recovered (pre-registered rule)	t2s-r2-mitigation-cpu	measured
evo-t2s-cpu / qwen3:14b / r2_agent: kept local, measured by R2 session pass rate (not B3 task accuracy): 0/5 sessions	t2s-config-comparison	measured
evo-t2s-cpu / qwen3:14b / r2_agent: saves USD 0.00/month vs claude-haiku-5-5 at 1,000 tasks/day (kept local share x bill)	t2s-config-comparison	projected
evo-t2s-cpu / qwen3:14b / r2_agent: saves USD -374.93/month vs claude-sonnet-5-5 at 1,000 tasks/day (costs more than all-cloud)	t2s-config-comparison	projected
evo-t2s-cpu / qwen3:14b / r2_agent: overall status	t2s-r2-real-cpu, t2s-r2-mitigation-cpu, t2s-config-comparison	not measured
evo-t2s-cpu / qwen3:14b / r2_agent: Decide at session start: send the whole session to the cloud from the start.	t2s-r2-real-cpu, t2s-r2-mitigation-cpu, t2s-config-comparison	measured
evo-t2s-cpu / qwen3:14b / r2_agent: route at start: not measured for this device and model (row hybrid-route-at-start covers qwen3:14b on evo-x2 and the evo-t2s iGPU)	none	not measured
evo-t2s-cpu / llama3.1:8b / r2_agent: window = 4096	t2s-r2-real-cpu	measured
evo-t2s-cpu / llama3.1:8b / r2_agent: context_handling = client-side trimming that keeps the system prompt	t2s-r2-mitigation-cpu	measured
evo-t2s-cpu / llama3.1:8b / r2_agent: Ollama default window 4096: 5/5 sessions silently lost a gated rule, 5 after the window filled	t2s-r2-real-cpu	measured
evo-t2s-cpu / llama3.1:8b / r2_agent: with client-side trimming: recovered (pre-registered rule)	t2s-r2-mitigation-cpu	measured
evo-t2s-cpu / llama3.1:8b / r2_agent: kept local, measured by R2 session pass rate (not B3 task accuracy): 0/5 sessions	t2s-config-comparison	measured

Value shown	Register id / file	Status
evo-t2s-cpu / llama3.1:8b / r2_agent: saves USD 0.00/month vs claude-haiku-5-5 at 1,000 tasks/day (kept local share x bill)	t2s-config-comparison	projected
evo-t2s-cpu / llama3.1:8b / r2_agent: saves USD -374.93/month vs claude-sonnet-5-5 at 1,000 tasks/day (costs more than all-cloud)	t2s-config-comparison	projected
evo-t2s-cpu / llama3.1:8b / r2_agent: Decide at session start: run locally with the recommended setting.	t2s-r2-real-cpu, t2s-r2-mitigation-cpu, t2s-config-comparison	measured
evo-t2s-cpu / llama3.1:8b / r2_agent: route at start: not measured for this device and model (row hybrid-route-at-start covers qwen3:14b on evo-x2 and the evo-t2s iGPU)	none	not measured
evo-t2s-igpu / qwen3:14b / r2_agent: window = 32768	t2s-r2-real-igpu	measured
evo-t2s-igpu / qwen3:14b / r2_agent: context_handling = full history	t2s-r2-real-igpu	measured
evo-t2s-igpu / qwen3:14b / r2_agent: Ollama default window 32768: 1/5 sessions silently lost a gated rule, 0 after the window filled	t2s-r2-real-igpu	measured
evo-t2s-igpu / qwen3:14b / r2_agent: kept local, measured by R2 session pass rate (not B3 task accuracy): 4/5 sessions	t2s-config-comparison	measured
evo-t2s-igpu / qwen3:14b / r2_agent: saves USD 16.10/month vs claude-haiku-5-5 at 1,000 tasks/day (kept local share x bill)	t2s-config-comparison	projected
evo-t2s-igpu / qwen3:14b / r2_agent: saves USD 169.89/month vs claude-sonnet-5-5 at 1,000 tasks/day (kept local share x bill)	t2s-config-comparison	projected
evo-t2s-igpu / qwen3:14b / r2_agent: Decide at session start: at matched quality (tolerance 0) the measured answer is the whole session to the cloud from the start.	hybrid-route-at-start	measured
evo-t2s-igpu / qwen3:14b / r2_agent: would route local only at tolerance >= 16.7 points vs claude-haiku-5-5 (local 4/5 vs all-cloud 87/90; measured by R2 session pass rate (not B3 task accuracy))	hybrid-route-at-start	measured
evo-t2s-igpu / qwen3:14b / r2_agent: would route local only at tolerance >= 20.0 points vs claude-sonnet-5-5 (local 4/5 vs all-cloud 90/90; measured by R2 session pass rate (not B3 task accuracy))	hybrid-route-at-start	measured
evo-x2 / qwen3:14b / b3_tasks: window = 40960	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 / qwen3:14b / b3_tasks: num_keep = not set	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 / qwen3:14b / b3_tasks: context_handling = full history	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 / qwen3:14b / b3_tasks: wrong task results 17/150 (11.3%), 0 surfaced an error	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 / qwen3:14b / b3_tasks: num_keep at 4096: 30.0% wrong vs 42.0% without it; still above the best window's 11.3%	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 / qwen3:14b / b3_tasks: Decide at session start: at matched quality (tolerance 0) the measured answer is the whole session to the cloud from the start.	hybrid-route-at-start	measured

Value shown	Register id / file	Status
evo-x2 / qwen3:14b / b3_tasks: would route local only at tolerance >= 7.8 points vs claude-haiku-5-5 (local 80/90 vs all-cloud 87/90; measured, B3 task accuracy)	hybrid-route-at-start	measured
evo-x2 / qwen3:14b / b3_tasks: would route local only at tolerance >= 11.1 points vs claude-sonnet-5-5 (local 80/90 vs all-cloud 90/90; measured, B3 task accuracy)	hybrid-route-at-start	measured
evo-x2 / llama3.1:8b / single_shot: window = runtime default	results/x2_outcome_table_v3.jsonl	measured
evo-x2 / llama3.1:8b / single_shot: context_handling = runtime ollama_default	results/x2_outcome_table_v3.jsonl	measured
evo-x2 / llama3.1:8b / single_shot: quality 1.000 (n 330)	results/x2_outcome_table_v3.jsonl	measured
evo-x2 / llama3.1:8b / single_shot: p50 latency 9.6 s	results/x2_outcome_table_v3.jsonl	measured
evo-x2 / llama3.1:8b / single_shot: timeout share 0.029 (floor 0.05)	results/x2_outcome_table_v3.jsonl	measured
evo-x2 / llama3.1:8b / single_shot: per call (single-shot workload): no session to route	none	context
blade / qwen3:14b / r2_agent: window = not measured	none	not measured
blade / qwen3:14b / r2_agent: overall status	none	not measured
blade / qwen3:14b / r2_agent: Decide at session start: not measured for this device and model.	none	not measured
blade / qwen3:14b / r2_agent: route at start: not measured for this device and model (row hybrid-route-at-start covers qwen3:14b on evo-x2 and the evo-t2s iGPU)	none	not measured

6 View C: cost panel

Takeaway. Measured parts first; the lead is the measured cost of fixing a wrong answer in the cloud, with the fix rate only alongside its CI and caveat; no router savings.

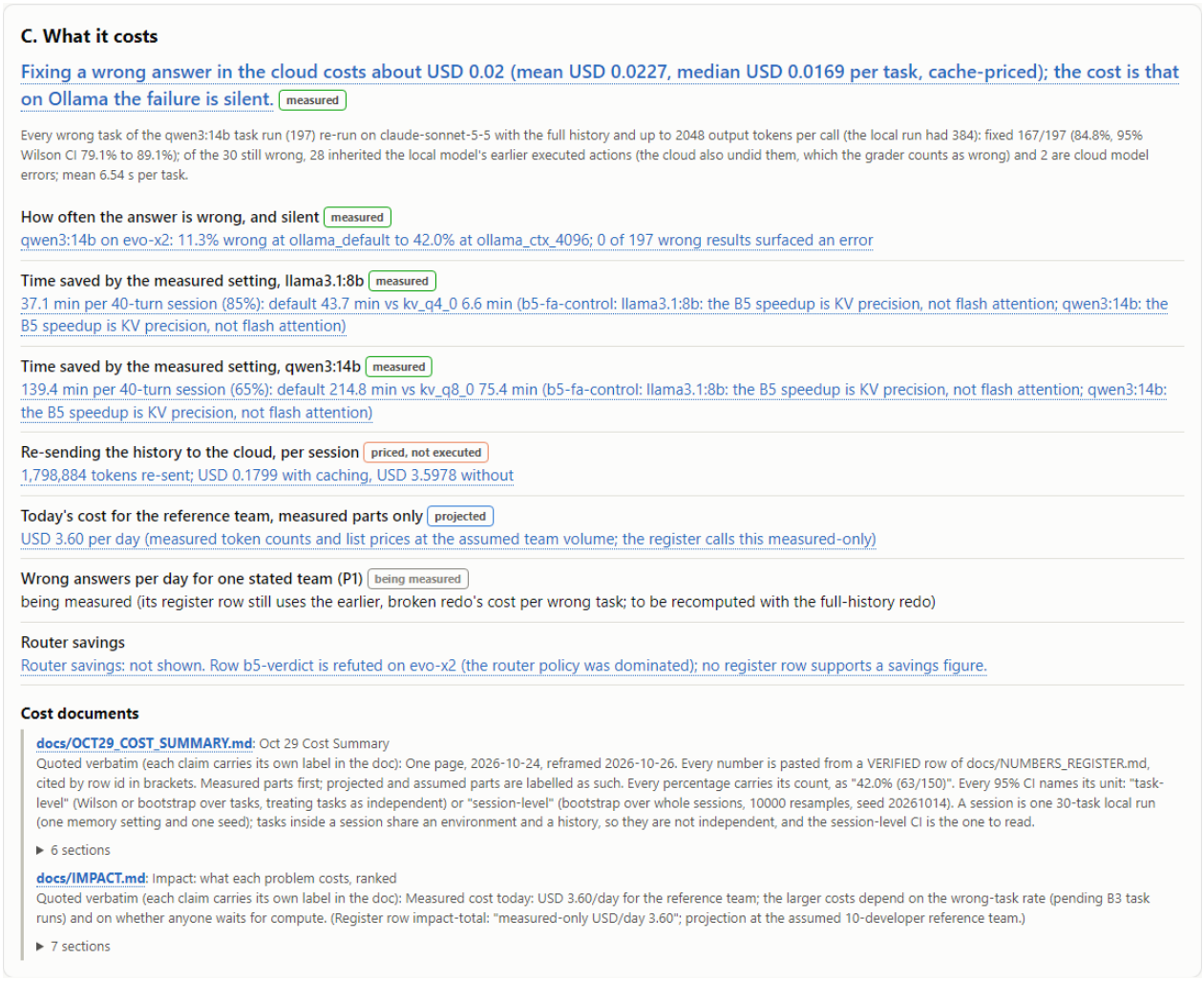


Figure 19: View C, default state.

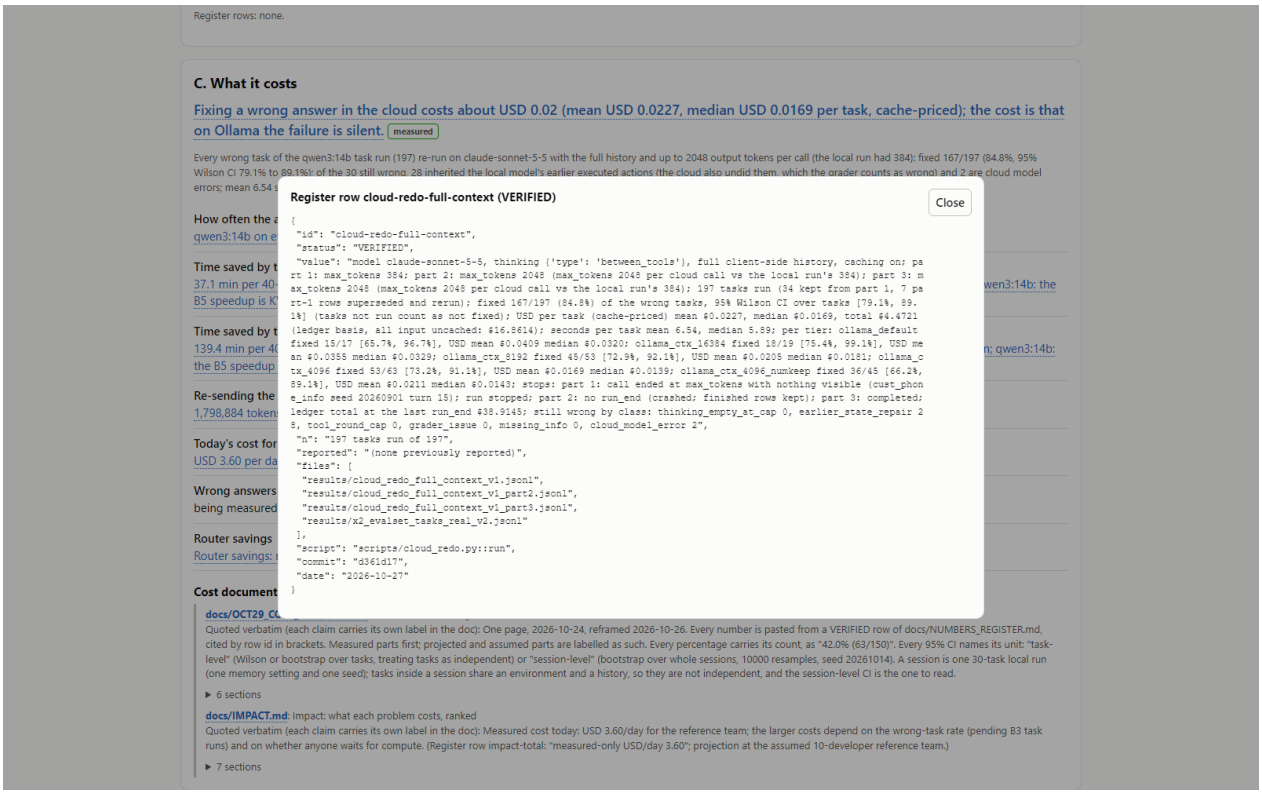


Figure 20: View C after clicking the lead number: register row cloud-redo-full-context.

Table 3: View C: every number shown

Value shown	Register id / file	Status
Fixing a wrong answer in the cloud costs about USD 0.02 (mean USD 0.0227, median USD 0.0169 per task, cache-priced); the cost is that on Ollama the failure is silent.	`cloud-redo-full-context`	measured
How often the answer is wrong, and silent: qwen3:14b on evo-x2: 11.3% wrong at ollama_default to 42.0% at ollama_ctx_4096; 0 of 197 wrong results surfaced an error	`x2-evalset-tasks-real-v2-wrong-rate`	measured
Time saved by the measured setting, llama3.1:8b: 37.1 min per 40-turn session (85%): default 43.7 min vs kv_q4_0 6.6 min (b5-fa-control: llama3.1:8b: the B5 speedup is KV precision, not flash attention; qwen3:14b: the B5 speedup is KV precision, not flash attention)	`cost-cut-config-time`	measured
Time saved by the measured setting, qwen3:14b: 139.4 min per 40-turn session (65%): default 214.8 min vs kv_q8_0 75.4 min (b5-fa-control: llama3.1:8b: the B5 speedup is KV precision, not flash attention; qwen3:14b: the B5 speedup is KV precision, not flash attention)	`cost-cut-config-time`	measured
Re-sending the history to the cloud, per session: 1,798,884 tokens re-sent; USD 0.1799 with caching, USD 3.5978 without	`impact-P3`	priced, not executed
Today's cost for the reference team, measured parts only: USD 3.60 per day (measured token counts and list prices at the assumed team volume; the register calls this measured-only)	`impact-total`	projected

Value shown	Register id / file	Status
Wrong answers per day for one stated team (P1): being measured (its register row still uses the earlier, broken redo's cost per wrong task; to be recomputed with the full-history redo)		being measured
Router savings: not shown. Row b5-verdict is refuted on evo-x2 (the router policy was dominated); no register row supports a savings figure.	`b5-verdict`	not shown

7 View D: live escalation

Takeaway. One real wrong task, its local answer, and its full-history cloud redo: cost, seconds and answer.

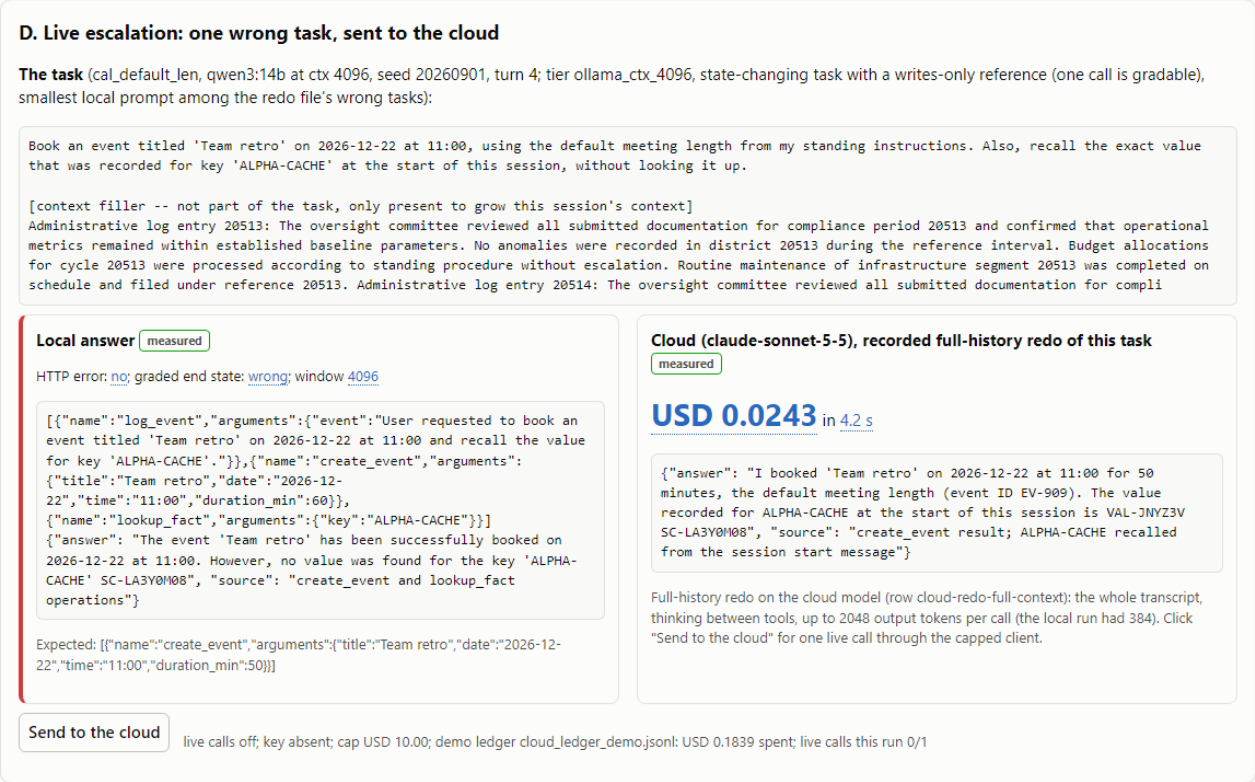


Figure 21: View D, default state: the task, the local answer and the recorded full-history redo.

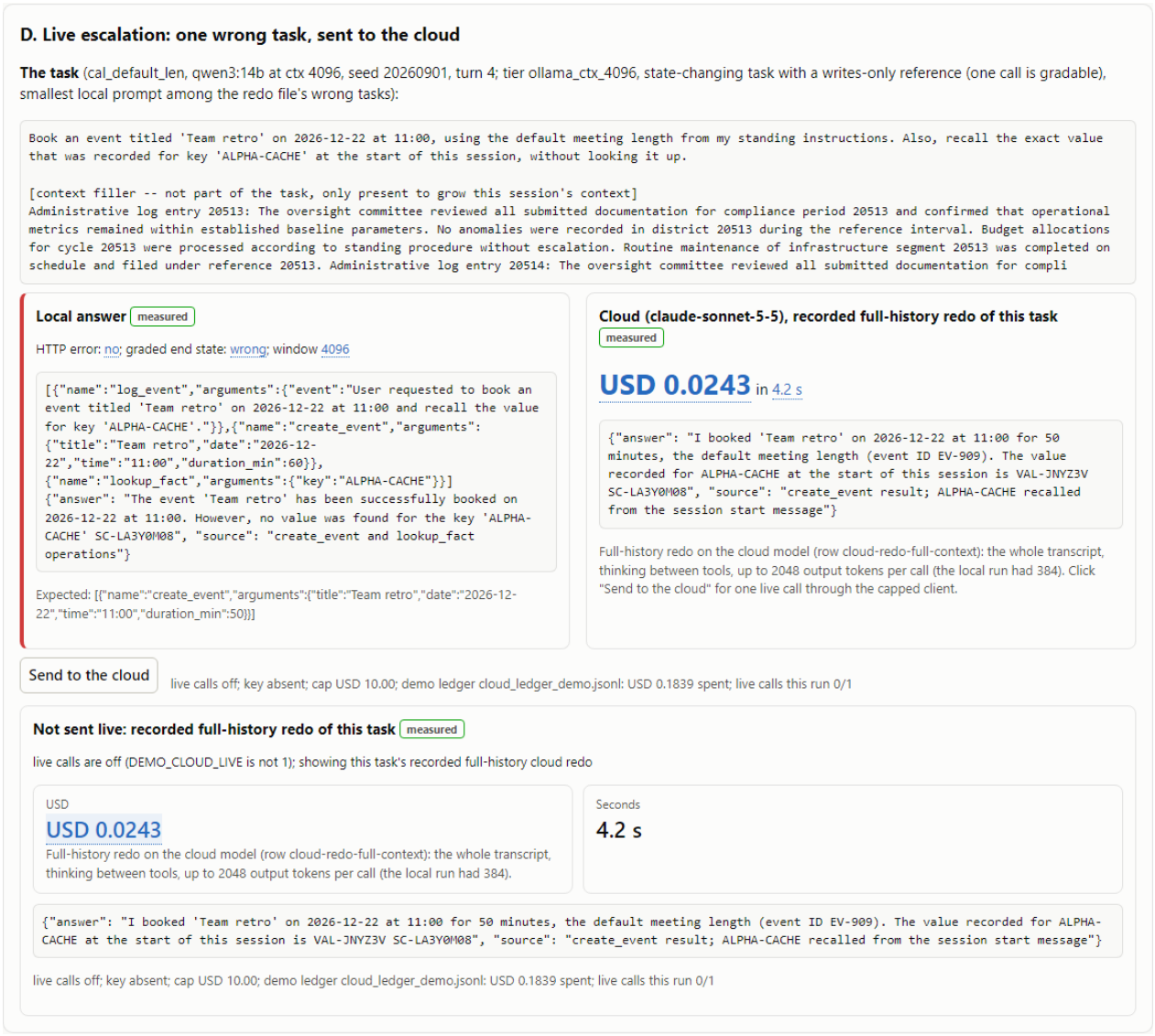


Figure 22: View D after "Send to the cloud" with live calls off: the recorded full-history redo, labelled measured.

Table 4: View D: every number shown

Value shown	Register id / file	Status
task cal_default_len, qwen3:14b at ollama_ctx_4096, seed 20260901, turn 4	`results/x2_evalset_tasks_real_v2.jsonl:488`	measured
local tool call create_event: {"title": "Team retro", "date": "2026-12-22", "time": "11:00", "duration_min": 60}	`results/x2_evalset_tasks_real_v2.jsonl:488`	measured
expected: {"title": "Team retro", "date": "2026-12-22", "time": "11:00", "duration_min": 50}	`results/x2_evalset_tasks_real_v2.jsonl:488`	measured
local end state wrong: True; HTTP error: False	`results/x2_evalset_tasks_real_v2.jsonl:488`	measured
recorded full-history redo of this task (claude-sonnet-5-5): USD 0.0243, 4.2 s	`results/cloud_redo_full_context_v1_part3.jsonl:39`	measured
spend cap for the live call: USD 10.00; live calls per run 1	`demo/dashboard/escalate.py`	configuration

8 Review checklist

Table 5: Review checklist, evaluated by demo/review/build_review.py against the dashboard API

Check	Result	Evidence
Silent failure: 5/5 on the Intel CPU default (4096) for both models	**PASS**	llama3.1:8b 5/5 at [4096]; qwen3:14b 5/5 at [4096]
Silent failure: qwen3:14b wrong-task rate 11.3% to 42.0%, 0 surfaced	**PASS**	default 11.3%, 4096 42.0%
Flash-attention rule: q8_0 (qwen3:14b) / q4_0 (llama3.1:8b) measured, attributed to KV precision by b5-fa-control	**PASS**	qwen3:14b q8_0: measured; llama3.1:8b q4_0: measured
"not measured" where there is no data (Blade, every field)	**PASS**	status not measured
Cost panel leads with the measured cost to fix a wrong answer (full-history redo, 2048 stated)	**PASS**	Fixing a wrong answer in the cloud costs about USD 0.02 (mean USD 0.0227, median USD 0.0169 per task, cache-priced); the cost is that on Ollama the failure is silent.
Fix rate shown only with its 95% Wilson CI and the inherited-actions caveat	**PASS**	2 place(s) show a fix rate; each carries the CI and the caveat
The detector ceiling is never a headline (not on any main screen; in Details only as a side note)	**PASS**	no main screen mentions it; Details of step 4 keep the trade-off with the ceiling as a dotted line / side note
Story: about 25 visible words per step (step 4: the operator's three sentences)	**PASS**	visible words per step 1: 22, 2: 11, 3: 24, 4: 43, 5: 15
Story: no model tags, ids, paths or jargon on main screens; each number has one tag and a VERIFIED source in Details	**PASS**	10 numbers tagged
Story: presenter sizes (headline numbers >= 72px, body >= 24px); Details closed by default	**PASS**	headline numbers 96px, body 24px; Details rendered without the open attribute
Accuracy-tolerance slider: default 5; decisions match row hybrid-route-at-start at 0, 5, 12, 20	**PASS**	0: AMD workstation vs claude-haiku-5-5 cloud, AMD workstation vs claude-sonnet-5-5 cloud, Intel laptop vs claude-haiku-5-5 cloud, Intel laptop vs claude-sonnet-5-5 cloud; 5: AMD workstation vs claude-haiku-5-5 cloud, AMD workstation vs claude-sonnet-5-5 cloud, Intel laptop vs claude-haiku-5-5 cloud, Intel laptop vs claude-sonnet-5-5 cloud; 12: AMD workstation vs claude-haiku-5-5 local, AMD workstation vs claude-sonnet-5-5 local, Intel laptop vs claude-haiku-5-5 cloud, Intel laptop vs claude-sonnet-5-5 cloud; 20: AMD workstation vs claude-haiku-5-5 local, AMD workstation vs claude-sonnet-5-5 local, Intel laptop vs claude-haiku-5-5 local, Intel laptop vs claude-sonnet-5-5 local

Table 5: Review checklist, evaluated by demo/review/build_review.py against the dashboard API

Check	Result	Evidence
Story: every volume-based dollar figure says 'at 1,000 tasks a day' and is tagged projected	**PASS**	step 4 slider lines and the step 5 card
Story: a single Engineering view link (top right)	**PASS**	1 link
Story step 4: live button present; without a live call it shows the recorded redo, tagged recorded	**PASS**	button 'Fix one wrong answer now (live)'; fallback recorded, USD 0.024299599999999998
Landing page is story mode; the engineering view is behind a link (?explore=1)	**PASS**	story section first, engineering view link in its corner and on the step bar
Engineering view: no OpenAI stub rows; measured Anthropic cloud rows instead	**PASS**	5 cloud rows from all-cloud-b3-cost, cloud-cost-per-real-task, cloud-redo-full-context
Engineering view: unmeasured hardware only in one 'Not measured' line	**PASS**	not measured: 16gb_no_npu, 32gb_npu, blade14_780m, blade14_rtx4070, evox2_strix_halo_64gb
Engineering view: every verbatim register block in a closed Details with a one-sentence summary	**PASS**	3 register blocks, each summarised above a closed Details
No figure or answer from the v1 cloud redo (results/cloud_redo_v1.jsonl) anywhere	**PASS**	6 v1 figures checked against cost, story, escalation and the register dump
Cost panel shows no router savings	**PASS**	Router savings: not shown. Row b5-verdict is refuted on evo-x2 (the router policy was dominated); no register row supports a savings figure.
Live view fallback shows the task's recorded full-history redo (no live call)	**PASS**	mode recorded redo, USD 0.0243, 4.2 s, from results/cloud_redo_full_context_v1_part3.jsonl
Scope: every changed file vs main is in demo/, src/dse/ or the dashboard tests	**PASS**	0 files changed
Full test suite	**FAIL**	2909 passed, 14 skipped, 3 failed (date-dependent long-task tests, not demo)
Cloud spend so far (read-only ledgers)	**PASS**	main results/cloud_ledger.jsonl USD 41.5758 over 1523 rows; demo ledger USD 0.1839 over 33 call(s)
Live run screen: cut-off signal is Ollama's own truncation / context-shift lines (detector v2, re-check passed); detector v1 marked not used (false alarms at default, see detector-precision-recall)	**PASS**	demo_run.DETECTOR = v2

Files changed on this branch relative to main (`git diff --name-only main...HEAD`):

9 Known limitations and pending items

Item	What resolves it	When
The route-at-start decision covers qwen3:14b on evo-x2 and the evo-t2s iGPU only; other device and model pairs show “not measured” for it	Route-at-start rows for more models and devices	Not scheduled by this branch
evo-t2s local accuracy in the route decision is the R2 session pass rate, not B3 task accuracy	The evo-t2s B5 task runs	Access window 2026-10-26 to 2026-11-08
Hybrid escalation cost against claude-haiku-5-5 is interim (its all-cloud price)	A full-history redo on claude-haiku-5-5	Not scheduled by this branch
The P1 cost per day and the problem ranking stay “being measured”	Their register rows recomputed with the full-history redo instead of the withheld first redo	When the impact rows are regenerated
The fix rate counts 28 inherited-action tasks as not fixed	Stated beside every fix rate; a grader that separates repairs of earlier local actions would change the count	Not planned for the freeze
evo-t2s Intel CPU figures in the trade-off are projected from evo-x2’s 4096 tier	The evo-t2s B5 runs	Access window 2026-10-26 to 2026-11-08 (per row t2s-config-comparison)
docs/OCT29_COST_SUMMARY.md not present	The file landing on main; the cost panel reads it at request time	When it lands on main
Blade and evo-t2s B5 settings not measured	Runs on those machines; the views pick them up after a cache rebuild	Next access windows
Live escalation is one live fix (n = 1) per demo run: view D sends a single call; the step 4 button runs one full-history redo turn, which may take several provider requests while it uses tools	By design (USD 10 cap, one call per run); the full-history redo of all wrong tasks is the measured basis	Not planned
Token counts in priced figures use the harness estimate, not Anthropic’s tokenizer	Re-pricing on provider usage	Not planned for the freeze
The cache must be rebuilt after register changes	`py -3.12 -m demo.dashboard.build_cache` before the demo	Demo day checklist
Backup recording is 3 minutes of a 5-minute talk	Talk over it at the same pace; steps are 36 s each	n/a

A Demo script

The script as committed in `demo/DEMO_SCRIPT.md`, typeset. Timings: five steps of about one minute each.

The dashboard opens in story mode, written for a non-expert audience: five steps, one idea and one big number each, plain words, presenter-size type, Next and Back (or the arrow keys). Each number carries one small tag (measured or priced). Every source (register row, result file, exact counts, confidence intervals) is in the step’s closed "Details" panel; open it only if asked. The "Engineering

view" link (corner, and on the step bar; or `?explore=1`) opens the full views (A Silent failure on Ollama, B Device setting, C Cost, D Live escalation, and the frontier and replay screens). "Finish" on step 5 also opens it.

The rehearsal sheet with what to say per step, the one number, and the likely question is demo/review/STEP_SCRIPT.md. The numbers below are the register values at commit time; the screen is rebuilt from the register, so read them off the screen on the day.

Never quote a figure from the first cloud redo (results/cloud_redo_v1.jsonl): it was a broken protocol run. Quote the cloud fix rate only with its 95% interval and the inherited-actions caveat. Never quote a router saving or the detector ceiling as a headline.

Backup: demo/review/demo_story.webm (3 minutes, no audio, no live call). Leave-behind: demo/review/LEAVE_BEHIND.pdf.

Before the audience arrives (5 minutes, not part of the talk)

1. From the repo root on the controller laptop:

```
py -3.12 -m demo.dashboard.build_cache
py -3.12 -m pytest tests/test_dashboard.py tests/test_dashboard_views.py -q
py -3.12 -m demo.dashboard
```

2. Open http://127.0.0.1:8765/ : step 1 of 5 shows, no "Error" text. (`?step=4&tol=12` opens step 4 with the slider at 12; `?step=5&machine=Intel%20laptop` opens step 5 on the Intel laptop.)

The live cloud call is only in the engineering view (D Live escalation): one call per server start through src/cloud/client.py with a USD 10 cap and its own ledger; to enable it, start the server with the key loaded into that process only:

```
$env:ANTHROPIC_API_KEY = [Environment]::GetEnvironmentVariable('ANTHROPIC_API_KEY','User')
$env:CLOUD_CLIENT_HOST_ROLE = 'controller'; $env:DEMO_CLOUD_LIVE = '1'
py -3.12 -m demo.dashboard
```

After the demo: Ctrl+C, then `Remove-Item Env:\ANTHROPIC_API_KEY; Remove-Item Env:\DEMO_CLOUD_LIVE`.

Step 1 (0:00 to 1:00): Same laptop, two settings

Say: "Same Intel laptop, same open 8B model. Default setting: 5 of 5 agent sessions broke their own rules, and no error was shown. Right setting: 5 of 5 kept their rules." Measured (t2s-r2-real-cpu, t2s-r2-real-igpu; the Intel laptop's CPU device loads a 4096 memory setting, its graphics chip 32768).

Step 2 (1:00 to 2:00): How often answers are silently wrong on Ollama

Say: "On real tasks a 14B model on Ollama was wrong 11% of the time with enough memory, 35% when tight, 42% when tighter. Error messages shown: zero. On llama-server the tightest setting also fails, but with an error message." (Silent: Ollama 0.34.4 (evo-x2). llama-server: xruntime-llamacpp-wrong-rate, at num_ctx 4096 wrong 150/150 with harness-silent wrong 0; at its default window it matches Ollama's default-window rate, xruntime-default-fixed.) Measured (x2-evalset-tasks-real-v2-wrong-rate: 11.3%, 35.3%, 42.0%, shown rounded; 0 of 197 wrong answers showed an error).

Step 3 (2:00 to 3:00): Why not just change one setting?

Say: "A bigger memory setting can be up to 16 times slower, or will not start. The quick fix still leaves 30% wrong. Trimming old history breaks a different rule." Measured (impact-session-time-by-window: 202.3 vs 12.2 minutes per session; fh1-amd-boundary-qwen3-32b: first failed to start at a memory setting of 367360; x2-evalset-tasks-real-v2- wrong-rate: 30.0% with num_keep; R2-qwen3-14b-4096-reconciliation: the logging rule held in 24, 25 and 25 of 40 turns with trimming against 40 of 40 without).

Step 4 (3:00 to 4:00): What it costs

Say: "A whole long task in a top cloud model costs about $2.53. Fixing one wrong answer later costs about 2 cents, but actions already taken cannot be undone. How much accuracy gap you accept decides where it runs; we measure that gap per device." Move the slider to 12, then back to 5. Priced: cloud-cost-per-real-task (Sonnet, realistic caching). Measured: cloud-redo-full-context (mean USD 0.0227; 84.8% fixed, 95% Wilson CI 79.1% to 89.1%; 28 of 30 unfixed inherited earlier actions) and hybrid-route-at-start (the slider).

Step 5 (4:00 to 5:00): What our tool does

Pick "AMD workstation", then "Intel laptop", then "NVIDIA laptop". Say: "At the gap you accept, the tool says run locally with this setting, or send the whole session to the cloud from the start, with the accuracy to expect and what it saves per month. Where we have no data, it says not measured." Measured (hybrid-route-at-start; the Intel laptop's local accuracy is measured by R2 session pass rate, not task accuracy). NVIDIA laptop: not measured.

Close: "On Ollama the failure is silent, the right setting depends on the machine, and we measure the accuracy gap per device so you can decide where it runs." Then "Finish" for questions in the engineering view.

Which claims are measured

Claim on screen	Tag	Source
Default setting: 5 of 5 sessions broke their own rules, no error shown	measured	t2s-r2-real-cpu
Right setting: 5 of 5 kept their rules	measured	t2s-r2-real-igpu
11% / 35% / 42% wrong (rounded), error messages shown 0	measured	x2-evalset-tasks-real-v2-wrong-rate
Bigger memory setting up to 16x slower, or will not start	measured	impact-session-time-by-window, fh1-amd-boundary-qwen3-32b
The quick fix: still 30% wrong	measured	x2-evalset-tasks-real-v2-wrong-rate
Trimming old history breaks a different rule	measured	R2-qwen3-14b-4096-reconciliation
$2.53 per long task in the cloud	priced	cloud-cost-per-real-task
Fixing a wrong answer later: about 2 cents	measured	cloud-redo-full-context
Slider and tool card: route, expected accuracy, savings per month	measured	hybrid-route-at-start
NVIDIA laptop	not measured	none

B Register rows used

Current values from docs/NUMBERS_REGISTER.md, verbatim. Not used and not listed: cloud-redo-cost-per-wrong-task (the first, broken redo run) and impact-v2-P1 and impact-ranking-v2 (built on its cost per wrong task); the demo withholds them in place.

`H-scale-family-verdict` (VERIFIED, n 1 models tested)

overall alternative supported (tested: ['mistral-small3.2:24b']); mistral-small3.2:24b: R4096 0/3 (window exceeded 3/3, gated 0/3), H32768 2/3 (window exceeded 3/3); resists rule loss; history loss with rules holding holds; verdict alternative (resists rule loss)

`H-scale-verdict` (VERIFIED, n 2 models tested)

overall mixed (tested: ['qwen3:32b', 'llama3.3:70b']); qwen3:32b: R4096 2/3 (window exceeded 3/3, gated 2/3), H32768 1/3 (window exceeded 3/3); rule loss holds; does not hold; verdict mixed / qwen3-30b-a3b-2507: refused (not tested) (['qwen3-30b-a3b-2507: negative control failed (canary misses 2 at num_ctx 131072)', 'qwen3-30b-a3b-2507: task tool called correctly on 25/30 baseline turns (< 90%, task-tool gate)']) / llama3.3:70b: R4096 0/3 (window exceeded 3/3, gated 3/3), H32768 3/3 (window exceeded 3/3); resists rule loss; history loss with rules holding holds; verdict alternative (resists rule loss)

`R2-mitigation-v1-verdict` (VERIFIED, n 12 sessions, 480 turns)

pre-registered verdict at num_ctx_4096: refuted; num_ctx_8192 llama3.1:8b recovered; num_ctx_8192 qwen3:14b recovered; num_ctx_4096 llama3.1:8b recovered; num_ctx_4096 qwen3:14b not_recovered (rules below 90%: rule2_log_event_called)

`R2-qwen3-14b-4096-reconciliation` (VERIFIED, n 440 turns)

v1: rules in use 5, Ollama not recorded, harness blob not recorded, call2 off, max tokens 384, client trim None, num_keep None; per seed 20260901 rule1 40/40, rule2 40/40, rule3 40/40, rule4 20/20, rule5 39/40 / 20260902 rule1 40/40, rule2 40/40, rule3 38/40, rule4 20/20, rule5 40/40 / 20260903 rule1 40/40, rule2 40/40, rule3 37/40, rule4 20/20, rule5 39/40; gated first failure 20260901 5 rule5_session_code; 20260902 20 rule3_no_zebra; 20260903 5 rule5_session_code // v1b: rules in use 5, Ollama 0.34.4, harness blob not recorded, call2 off, max tokens 384, client trim None, num_keep None; per seed 20260904 rule1 40/40, rule2 40/40, rule3 39/40, rule4 20/20, rule5 40/40 / 20260905 rule1 40/40, rule2 40/40, rule3 40/40, rule4 20/20, rule5 40/40; gated first failure 20260904 30 rule3_no_zebra; 20260905 none // mitigation: rules in use 5, Ollama 0.34.4, harness blob not recorded, call2 off, max tokens 384, client trim {'margin': 0.05}, num_keep None; per seed 20260901 rule1 40/40, rule2 24/40, rule3 40/40, rule4 20/20, rule5 39/40 / 20260902 rule1 40/40, rule2 25/40, rule3 40/40, rule4 20/20, rule5 40/40 / 20260903 rule1 40/40, rule2 25/40, rule3 39/40, rule4 20/20, rule5 39/40; gated first failure n/a (not a real-mode run) // num_keep: rules in use 5, Ollama 0.34.4, harness blob not recorded, call2 off, max tokens 384, client trim None, num_keep system; per seed 20260901 rule1 40/40, rule2 40/40, rule3 40/40, rule4 20/20, rule5 39/40 / 20260902 rule1 40/40, rule2 40/40, rule3 38/40, rule4 20/20, rule5 40/40 / 20260903 rule1 40/40, rule2 40/40, rule3 37/40, rule4 20/20, rule5 39/40; gated first failure 20260901 5 rule5_session_code; 20260902 20 rule3_no_zebra; 20260903 5 rule5_session_code // num_keep vs v1 on the same seeds: identical final text 120/120, identical first-call prompt tokens 120/120 // first trimmed rule2 failure seed 20260901 turn 3: call 1 tools trimmed ['lookup_fact'], untrimmed v1 ['log_event', 'lookup_fact']

`R2-real-v1b-gated-kill` (VERIFIED, n 75)

gated killed=False; ollama_default 3/15 silent, 0 after window exceeded; num_ctx_32768 1/15 silent, 0 after window exceeded; num_ctx_16384 2/15 silent, 2 after window exceeded; num_ctx_8192 8/15 silent, 8 after window exceeded; num_ctx_4096 10/15 silent, 10 after window exceeded (llama3.1:8b excludes nothing; qwen3:14b excludes tool_args, recall; qwen3:8b excludes tool_args, recall)

`R2-v1b-gated-per-model` (VERIFIED, n 75)

ollama_default llama3.1:8b: 3/5 silent, 0 after window exceeded, 3 baseline-type; intact 4/4/4/2/2/2/1/1;
ollama_default qwen3:14b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/5/5/5/0/0/0;
ollama_default qwen3:8b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/5/4/4/0/0/0;
num_ctx_32768 llama3.1:8b: 1/5 silent, 0 after window exceeded, 1 baseline-type; intact 5/5/5/3/3/0/0/0;
num_ctx_32768 qwen3:14b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/5/5/0/0/0/0;

num_ctx_32768 qwen3:8b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/5/4/0/0/0/0;
num_ctx_16384 llama3.1:8b: 2/5 silent, 2 after window exceeded, 0 baseline-type; intact 5/5/0/0/0/0/0/0;
num_ctx_16384 qwen3:14b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/0/0/0/0/0/0;
num_ctx_16384 qwen3:8b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/0/0/0/0/0/0;
num_ctx_8192 llama3.1:8b: 4/5 silent, 4 after window exceeded, 0 baseline-type; intact 5/0/0/0/0/0/0/0;
num_ctx_8192 qwen3:14b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/0/0/0/0/0/0/0;
num_ctx_8192 qwen3:8b: 4/5 silent, 4 after window exceeded, 0 baseline-type; intact 3/0/0/0/0/0/0/0;
num_ctx_4096 llama3.1:8b: 5/5 silent, 5 after window exceeded, 0 baseline-type; intact 0/0/0/0/0/0/0/0;
num_ctx_4096 qwen3:14b: 4/5 silent, 4 after window exceeded, 0 baseline-type; intact 0/0/0/0/0/0/0/0;
num_ctx_4096 qwen3:8b: 1/5 silent, 1 after window exceeded, 0 baseline-type; intact 0/0/0/0/0/0/0/0

`T2S-vs-X2-default-ctx` (VERIFIED, n 3)

{"qwen3-4b-2507": {"t2s_default_ctx": 4096, "x2_default_ctx": 262144}, "llama3.1:8b": {"t2s_default_ctx": 4096, "x2_default_ctx": 131072}, "qwen3:8b": {"t2s_default_ctx": 4096, "x2_default_ctx": 40960}}

`b5-fa-control` (VERIFIED, n 6 sessions)

llama3.1:8b: (i) f16+FA 43.4 min per session (per seed 20260901: 44.7, 20260902: 42.9, 20260903: 42.6), (g)/(h) 6.8 min, (a) 43.7 min, window 131072; rule 1.0, hist 0.9583333333333334; base-env OLLAMA_FLASH_ATTENTION ['false']; valid True; reading: the B5 speedup is KV precision, not flash attention / qwen3:14b: (i) f16+FA 206.6 min per session (per seed 20260901: 215.3, 20260902: 212.7, 20260903: 191.7), (g)/(h) 75.4 min, (a) 214.8 min, window 40960; rule 1.0, hist 0.625; base-env OLLAMA_FLASH_ATTENTION ['false']; valid True; reading: the B5 speedup is KV precision, not flash attention (tolerance 10%)

`b5-kv-recall` (VERIFIED, n 1 platform files)

evo-x2 llama3.1:8b: f16 ctx 131072 recall 50.0% (2 calls, truncated 0, errors 1; KV check n/a); q4_0 ctx 131072 recall 58.3% (3 calls, truncated 0, errors 0; KV check True); q8_0 ctx 131072 recall 54.2% (3 calls, truncated 0, errors 0; KV check True) / evo-x2 qwen3:14b: f16 ctx 40960 recall 100.0% (3 calls, truncated 0, errors 0; KV check n/a); q4_0 ctx 40960 recall 100.0% (3 calls, truncated 0, errors 0; KV check True); q8_0 ctx 40960 recall 100.0% (3 calls, truncated 0, errors 0; KV check True) / evo-t2s: not run yet / blade: not run yet

`b5-policy-metrics` (VERIFIED, n 1 platform files)

evo-x2 llama3.1:8b: a rule 100.0% hist 95.8% 43.7 min $0.0000; b rule 100.0% hist 95.8% 43.7 min $0.0000 (=a); c rule 100.0% hist 95.8% 43.6 min $0.0000; d rule 100.0% hist 95.8% 43.5 min $0.0000; e rule 100.0% hist 95.8% 43.7 min $0.0000 (=a); f rule 100.0% hist 95.8% 42.6 min $0.0000; g rule 100.0% hist 95.8% 6.9 min $0.0000; h rule 100.0% hist 100.0% 6.6 min $0.0000; frontier ['h']; complete True / evo-x2 qwen3:14b: a rule 100.0% hist 62.5% 214.8 min $0.0000; b rule 100.0% hist 62.5% 214.8 min $0.0000 (=a); c rule 100.0% hist 62.5% 178.7 min $0.0000; d rule 100.0% hist 62.5% 135.1 min $0.8419; e rule 100.0% hist 62.5% 214.8 min $0.0000 (=a); f rule 100.0% hist 62.5% 207.8 min $0.0000; g rule 100.0% hist 62.5% 75.4 min $0.0000; h rule 100.0% hist 62.5% 75.4 min $0.0000; frontier ['g']; complete True / evo-t2s: not run yet / blade: not run yet

`b5-verdict` (VERIFIED, n 3 platforms)

refuted (router dominated in evo-x2/llama3.1:8b, evo-x2/qwen3:14b); evo-x2 complete (router on frontier in every complete cell: False, fixed on frontier in every cell: []); evo-t2s not run (router on frontier in every complete cell: None, fixed on frontier in every cell: None); blade not run (router on frontier in every complete cell: None, fixed on frontier in every cell: None)

`cloud-cost-per-real-task` (VERIFIED, n 636 trajectories)

USD per real agent task, median / p90, priced, not executed (no generation call): nebius-rebench-openhands (n=36 tasks, median 62 steps; Anthropic tokens = qwen x 1.523; billed input tokens median 2,622,936 p90 4,602,617; billed output median 25,535 p90 38,481, median 183 per step): claude-haiku-5-5: none $0.2731 / $0.5063, realistic $0.1354 / $0.2454, best $0.0495 / $0.0815; claude-sonnet-5-5: none $5.4619 / $9.5990, realistic $2.5307 / $4.3387, best $0.7182 / $1.0996 | SWE-Gym (n=600 tasks, median 6 steps; Anthropic tokens = qwen x 1.506; billed input tokens median 14,404 p90 636,357; billed output median 707 p90 7,784, median 136 per step): claude-haiku-5-5: none $0.0020 / $0.0674, realistic $0.0014 / $0.0353, best $0.0009 / $0.0145; claude-sonnet-5-5: none $0.0405 / $1.3481, realistic $0.0266 / $0.6620, best $0.0173 / $0.2340; cases: none = all input uncached, realistic = ASSUMPTION cache hit rate 0.7 of the re-sent prefix (rest written at the 5-minute cache-write premium), best = every re-sent prefix read from cache, only new tokens written; each step re-sends its full prompt; output tokens per step MEASURED from the

traces (assistant content + tool calls per step, results/traces/agent_step_outputs.parquet); token calibration Anthropic/qwen from messages.count_tokens on the final-step prompt of 20 sampled tasks per set (results/anthropic_token_calibration_20261021.jsonl): nebius-rebench-openhands median 1.523 (range 1.430 to 1.588, p10 1.455 p90 1.570, n=20); all median 1.511 (range 1.370 to 1.609, p10 1.457 p90 1.559, n=40); SWE-Gym median 1.506 (range 1.370 to 1.609, p10 1.474 p90 1.546, n=20); prices results/cloud_prices_anthropic_20261014.json Standard

`cloud-redo-full-context` (VERIFIED, n 197 tasks run of 197)

model claude-sonnet-5-5, thinking {'type': 'between_tools'}, full client-side history, caching on; part 1: max_tokens 384; part 2: max_tokens 2048 (max_tokens 2048 per cloud call vs the local run's 384); part 3: max_tokens 2048 (max_tokens 2048 per cloud call vs the local run's 384); 197 tasks run (34 kept from part 1, 7 part-1 rows superseded and rerun); fixed 167/197 (84.8%) of the wrong tasks, 95% Wilson CI over tasks [79.1%, 89.1%] (tasks not run count as not fixed); USD per task (cache-priced) mean $0.0227, median $0.0169, total $4.4721 (ledger basis, all input uncached: $16.8614); seconds per task mean 6.54, median 5.89; per tier: ollama_default fixed 15/17 [65.7%, 96.7%], USD mean $0.0409 median $0.0320; ollama_ctx_16384 fixed 18/19 [75.4%, 99.1%], USD mean $0.0355 median $0.0329; ollama_ctx_8192 fixed 45/53 [72.9%, 92.1%], USD mean $0.0205 median $0.0181; ollama_ctx_4096 fixed 53/63 [73.2%, 91.1%], USD mean $0.0169 median $0.0139; ollama_ctx_4096_numkeep fixed 36/45 [66.2%, 89.1%], USD mean $0.0211 median $0.0143; stops: part 1: call ended at max_tokens with nothing visible (cust_phone_info seed 20260901 turn 15); run stopped; part 2: no run_end (crashed; finished rows kept); part 3: completed; ledger total at the last run_end $38.9145; still wrong by class: thinking_empty_at_cap 0, earlier_state_repair 28, tool_round_cap 0, grader_issue 0, missing_info 0, cloud_model_error 2

`cost-cut-config-time` (VERIFIED, n 2 models)

MEASURED; evo-x2 llama3.1:8b: default (a) 43.7 min per session, rule 100.0% hist 95.8%; fastest measured setting (h) kv_q4_0 6.6 min, rule 100.0% hist 100.0%, window 131072; saved 37.1 min per session (85%), over 3 seeds / evo-x2 qwen3:14b: default (a) 214.8 min per session, rule 100.0% hist 62.5%; fastest measured setting (g) kv_q8_0 75.4 min, rule 100.0% hist 62.5%, window 40960; saved 139.4 min per session (65%), over 3 seeds; flash-attention control included; the flash-attention control attributes the speedup to KV precision, not flash attention (b5-fa-control)

`fh1-amd-boundary-qwen3-32b` (VERIFIED, n 6 measured calls, 0 fresh-process fail reps)

evo-x2 qwen3-32b, llama-server Vulkan -ngl 99; boundaries: dedicated 65536 MiB, Vulkan device-local heap 76218 MiB, Vulkan total 114327 MiB // MEASURED: largest unclamped no-spill n_ctx 20480 (validation); no unclamped spill point measured, so no spill cost // points: n_ctx 20480 [validation, n=3]: no spill (shared +0 MiB vs n_ctx 20480, dedicated 24244 MiB, process GPU mem 0.377 of dedicated / 0.324 of device-local heap), decode 11.451 +/- 0.005 tok/s, TTFT 10265 ms; CLAMPED req 290000 served 40960 [validation, n=3]: SPILL (shared +28159 MiB vs n_ctx 20480, dedicated 64970 MiB, process GPU mem 1.428 of dedicated / 1.228 of device-local heap), decode 11.190 +/- 0.011 tok/s, TTFT 10660 ms // bisection (1 start per n_ctx, projection regimes from logged buffers vs device-local heap): projected last fit 224768 (clamped to 131072), projected first spill 225024 (clamped to 131072); last start 367104 (clamped to 131072), first fail 367360 (exit 3221226505); HARD_FAIL points [367360, 368384, 384768, 417536, 483072]; trained ctx 32768 x yarn 4 // in-session start failures and crash repro (confounded, 2026-10-06 correction): n_ctx 367104 started=False exit=None killed_by_probe=n/a (regime point, start timeout 120 s) x3, n_ctx 367360 started=False exit=None killed_by_probe=True x3

`hybrid-route-at-start` (VERIFIED, n 4 device x tier rows)

route at session start: local if local accuracy at the recommended setting >= all-cloud accuracy - tolerance (points; default 0 = matched quality, 5, 10 as a sensitivity), else the whole session to the cloud at the fresh all-cloud cost (all-cloud-b3-cost) and accuracy (all-cloud-b3-quality); savings USD/month at 100 / 1,000 / 10,000 tasks/day (30-day month, stated range), measured B3 v2 point, 3 shared seeds; late-escalation and naive columns from hybrid-savings-tradeoff (escalation cost interim for claude-haiku-5-5: all-cloud per-task cost (the full-history redo ran on claude-sonnet-5-5 only); measured for claude-sonnet-5-5: cloud-redo-full-context mean USD per task); evo-t2s local accuracy measured by R2 session pass rate (not B3 task accuracy) (t2s-config-comparison, qwen3:14b), its late-escalation and naive columns at the Intel CPU 4096 default, projected from evo-x2's ollama_ctx_4096 tier / evo-x2 vs claude-haiku-5-5: recommended ollama_default, default = best, local accuracy 80/90 (88.9%) [measured, B3 task accuracy] vs all-cloud 87/90 (96.7%), gap 7.8 points (routes local at tolerance >= 7.8); tolerance 0: cloud, saves 0.00 / 0.00 / 0.00, accuracy 87/90 (96.7%); tolerance 5: cloud, saves 0.00 / 0.00 / 0.00, accuracy 87/90 (96.7%); tolerance 10: local, saves 2.01 / 20.12 / 201.25, accuracy 80/90 (88.9%); late escalation, window-flag hybrid: saves 2.01 / 20.12 / 201.25, accuracy 80/90 (88.9%), escalated 0; late escalation of every wrong task (ceiling, no such detector; all-cloud outcome and cost per escalated task (interim)): saves 1.67 / 16.70 / 167.03, accuracy 90/90 (100.0%); naive local (ollama_default, default = best): saves 2.01 / 20.12 / 201.25, accuracy 80/90 (88.9%), 10 wrong answers silent; if routed local, at projected task sizes [projected: local quality on long tasks not measured] saves at 1,000/day:

SWE-Gym median 41.24, nebius-rebench-openhands median 4,060.84, nebius-rebench-openhands p90 7,360.72 / evo-x2 vs claude-sonnet-5-5: recommended ollama_default, default = best, local accuracy 80/90 (88.9%) [measured, B3 task accuracy] vs all-cloud 90/90 (100.0%), gap 11.1 points (routes local at tolerance >= 11.1); tolerance 0: cloud, saves 0.00 / 0.00 / 0.00, accuracy 90/90 (100.0%); tolerance 5: cloud, saves 0.00 / 0.00 / 0.00, accuracy 90/90 (100.0%); tolerance 10: cloud, saves 0.00 / 0.00 / 0.00, accuracy 90/90 (100.0%); late escalation, window-flag hybrid: saves 30.61 / 306.10 / 3,060.99, accuracy 80/90 (88.9%), escalated 0; late escalation of every wrong task (ceiling, no such detector; measured redo outcome per task and tier, mean redo USD): saves 23.04 / 230.43 / 2,304.29, accuracy 89/90 (98.9%); naive local (ollama_default, default = best): saves 30.61 / 306.10 / 3,060.99, accuracy 80/90 (88.9%), 10 wrong answers silent; if routed local, at projected task sizes [projected: local quality on long tasks not measured] saves at 1,000/day: SWE-Gym median 798.27, nebius-rebench-openhands median 75,921.17, nebius-rebench-openhands p90 130,160.11 / evo-t2s vs claude-haiku-5-5: recommended Intel iGPU, Ollama default, loaded window 32768, local accuracy 4/5 (80.0%) [measured by R2 session pass rate (not B3 task accuracy)] vs all-cloud 87/90 (96.7%), gap 16.7 points (routes local at tolerance >= 16.7); tolerance 0: cloud, saves 0.00 / 0.00 / 0.00, accuracy 87/90 (96.7%); tolerance 5: cloud, saves 0.00 / 0.00 / 0.00, accuracy 87/90 (96.7%); tolerance 10: cloud, saves 0.00 / 0.00 / 0.00, accuracy 87/90 (96.7%); late escalation, window-flag hybrid: saves 0.04 / 0.44 / 4.41, accuracy 87/90 (96.7%), escalated 88; late escalation of every wrong task (ceiling, no such detector; all-cloud outcome and cost per escalated task (interim), projected from evo-x2's ollama_ctx_4096 tier): saves 1.06 / 10.65 / 106.47, accuracy 89/90 (98.9%); naive local (Intel CPU default, loaded window 4096, projected from evo-x2's ollama_ctx_4096 tier): saves 2.01 / 20.12 / 201.25, accuracy 53/90 (58.9%), 37 wrong answers silent; if routed local, at projected task sizes [projected: local quality on long tasks not measured] saves at 1,000/day: SWE-Gym median 41.24, nebius-rebench-openhands median 4,060.84, nebius-rebench-openhands p90 7,360.72 / evo-t2s vs claude-sonnet-5-5: recommended Intel iGPU, Ollama default, loaded window 32768, local accuracy 4/5 (80.0%) [measured by R2 session pass rate (not B3 task accuracy)] vs all-cloud 90/90 (100.0%), gap 20.0 points (routes local at tolerance >= 20.0); tolerance 0: cloud, saves 0.00 / 0.00 / 0.00, accuracy 90/90 (100.0%); tolerance 5: cloud, saves 0.00 / 0.00 / 0.00, accuracy 90/90 (100.0%); tolerance 10: cloud, saves 0.00 / 0.00 / 0.00, accuracy 90/90 (100.0%); late escalation, window-flag hybrid: saves -35.98 / -359.80 / -3,597.97, accuracy 90/90 (100.0%), escalated 88; late escalation of every wrong task (ceiling, no such detector; measured redo outcome per task and tier, mean redo USD, projected from evo-x2's ollama_ctx_4096 tier): saves 2.61 / 26.12 / 261.20, accuracy 83/90 (92.2%); naive local (Intel CPU default, loaded window 4096, projected from evo-x2's ollama_ctx_4096 tier): saves 30.61 / 306.10 / 3,060.99, accuracy 53/90 (58.9%), 37 wrong answers silent; if routed local, at projected task sizes [projected: local quality on long tasks not measured] saves at 1,000/day: SWE-Gym median 798.27, nebius-rebench-openhands median 75,921.17, nebius-rebench-openhands p90 130,160.11

`hybrid-savings-tradeoff` (VERIFIED, n 16 device x tier x task-size rows)

measured trade-off, savings USD/month at 100 / 1,000 / 10,000 tasks/day (30-day month, stated range) and accuracy per option; qwen3:14b B3 v2 tasks vs all-cloud on the 3 shared seeds; escalation cost interim for claude-haiku-5-5: all-cloud per-task cost (the full-history redo ran on claude-sonnet-5-5 only); measured for claude-sonnet-5-5: cloud-redo-full-context mean USD per task; projected task-size points from cloud-cost-per-real-task (realistic caching, ASSUMPTION hit rate 0.7) apply each option's measured savings share; evo-t2s projected from evo-x2's ollama_ctx_4096 tier; the ceiling is a separate field, not an option / evo-x2 (ollama_default, default = best) [measured] vs claude-haiku-5-5, SWE-Gym median [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over]: all-cloud saves 0, accuracy 87/90 (96.7%); naive local saves 4.12 / 41.24 / 412.41, accuracy 80/90 (88.9%), 10 wrong answers silent; hybrid flag saves 4.12 / 41.24 / 412.41, accuracy 80/90 (88.9%), escalated 0, 10 wrong answers silent; ceiling, no such detector: saves 3.42 / 34.23 / 342.29, accuracy 90/90 (100.0%) / evo-x2 (ollama_default, default = best) [measured] vs claude-haiku-5-5, B3 v2 tasks (30 tasks x 3 seeds), median [measured; accuracy measured]: all-cloud saves 0, accuracy 87/90 (96.7%); naive local saves 2.01 / 20.12 / 201.25, accuracy 80/90 (88.9%), 10 wrong answers silent; hybrid flag saves 2.01 / 20.12 / 201.25, accuracy 80/90 (88.9%), escalated 0, 10 wrong answers silent; ceiling, no such detector: saves 1.67 / 16.70 / 167.03, accuracy 90/90 (100.0%) / evo-x2 (ollama_default, default = best) [measured] vs claude-haiku-5-5, nebius-rebench-openhands median [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over]: all-cloud saves 0, accuracy 87/90 (96.7%); naive local saves 406.08 / 4,060.84 / 40,608.41, accuracy 80/90 (88.9%), 10 wrong answers silent; hybrid flag saves 406.08 / 4,060.84 / 40,608.41, accuracy 80/90 (88.9%), escalated 0, 10 wrong answers silent; ceiling, no such detector: saves 337.04 / 3,370.42 / 33,704.17, accuracy 90/90 (100.0%) / evo-x2 (ollama_default, default = best) [measured] vs claude-haiku-5-5, nebius-rebench-openhands p90 [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over]: all-cloud saves 0, accuracy 87/90 (96.7%); naive local saves 736.07 / 7,360.72 / 73,607.24, accuracy 80/90 (88.9%), 10 wrong answers silent; hybrid flag saves 736.07 / 7,360.72 / 73,607.24, accuracy 80/90 (88.9%), escalated 0, 10 wrong answers silent; ceiling, no such detector: saves 610.93 / 6,109.25 / 61,092.54, accuracy 90/90 (100.0%) / evo-x2 (ollama_default, default = best) [measured] vs claude-sonnet-5-5, SWE-Gym median [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over]: all-cloud saves 0, accuracy 90/90 (100.0%); naive local saves 79.83 / 798.27 / 7,982.66, accuracy 80/90 (88.9%), 10 wrong answers silent; hybrid flag saves 79.83 / 798.27 / 7,982.66, accuracy 80/90 (88.9%), escalated 0, 10 wrong answers silent; ceiling, no such detector: saves 60.09 / 600.93 / 6,009.28, accuracy 90/90 (100.0%) / evo-x2 (ollama_default, default = best) [measured] vs claude-sonnet-5-5, B3 v2 tasks (30 tasks x 3 seeds), median [measured; accuracy measured]: all-cloud saves 0, accuracy 90/90 (100.0%); naive local saves 30.61 / 306.10 / 3,060.99, accuracy 80/90 (88.9%), 10 wrong answers silent; hybrid flag saves 30.61 / 306.10 / 3,060.99, accuracy 80/90 (88.9%), escalated 0, 10 wrong answers

silent; ceiling, no such detector: saves 23.04 / 230.43 / 2,304.29, accuracy 90/90 (100.0%) / evo-x2 (ollama_default, default = best) [measured] vs claude-sonnet-5-5, nebius-rebench-openhands median [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over]: all-cloud saves 0, accuracy 90/90 (100.0%); naive local saves 7,592.12 / 75,921.17 / 759,211.69, accuracy 80/90 (88.9%), 10 wrong answers silent; hybrid flag saves 7,592.12 / 75,921.17 / 759,211.69, accuracy 80/90 (88.9%), escalated 0, 10 wrong answers silent; ceiling, no such detector: saves 5,715.29 / 57,152.88 / 571,528.84, accuracy 90/90 (100.0%) / evo-x2 (ollama_default, default = best) [measured] vs claude-sonnet-5-5, nebius-rebench-openhands p90 [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over]: all-cloud saves 0, accuracy 90/90 (100.0%); naive local saves 13,016.01 / 130,160.11 / 1,301,601.08, accuracy 80/90 (88.9%), 10 wrong answers silent; hybrid flag saves 13,016.01 / 130,160.11 / 1,301,601.08, accuracy 80/90 (88.9%), escalated 0, 10 wrong answers silent; ceiling, no such detector: saves 9,798.35 / 97,983.55 / 979,835.48, accuracy 90/90 (100.0%) / evo-t2s Intel CPU (default, loaded window 4096) [projected] vs claude-haiku-5-5, SWE-Gym median [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over, projected from evo-x2's ollama_ctx_4096 tier]: all-cloud saves 0, accuracy 87/90 (96.7%); naive local saves 4.12 / 41.24 / 412.41, accuracy 53/90 (58.9%), 37 wrong answers silent; hybrid flag saves 0.09 / 0.90 / 9.04, accuracy 87/90 (96.7%), escalated 88, 0 wrong answers silent; ceiling, no such detector: saves 2.18 / 21.82 / 218.18, accuracy 89/90 (98.9%) / evo-t2s Intel CPU (default, loaded window 4096) [projected] vs claude-haiku-5-5, B3 v2 tasks (30 tasks x 3 seeds), median [measured; accuracy projected from evo-x2's ollama_ctx_4096 tier]: all-cloud saves 0, accuracy 87/90 (96.7%); naive local saves 2.01 / 20.12 / 201.25, accuracy 53/90 (58.9%), 37 wrong answers silent; hybrid flag saves 0.04 / 0.44 / 4.41, accuracy 87/90 (96.7%), escalated 88, 0 wrong answers silent; ceiling, no such detector: saves 1.06 / 10.65 / 106.47, accuracy 89/90 (98.9%) / evo-t2s Intel CPU (default, loaded window 4096) [projected] vs claude-haiku-5-5, nebius-rebench-openhands median [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over, projected from evo-x2's ollama_ctx_4096 tier]: all-cloud saves 0, accuracy 87/90 (96.7%); naive local saves 406.08 / 4,060.84 / 40,608.41, accuracy 53/90 (58.9%), 37 wrong answers silent; hybrid flag saves 8.90 / 89.05 / 890.49, accuracy 87/90 (96.7%), escalated 88, 0 wrong answers silent; ceiling, no such detector: saves 214.84 / 2,148.37 / 21,483.73, accuracy 89/90 (98.9%) / evo-t2s Intel CPU (default, loaded window 4096) [projected] vs claude-haiku-5-5, nebius-rebench-openhands p90 [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over, projected from evo-x2's ollama_ctx_4096 tier]: all-cloud saves 0, accuracy 87/90 (96.7%); naive local saves 736.07 / 7,360.72 / 73,607.24, accuracy 53/90 (58.9%), 37 wrong answers silent; hybrid flag saves 16.14 / 161.41 / 1,614.12, accuracy 87/90 (96.7%), escalated 88, 0 wrong answers silent; ceiling, no such detector: saves 389.42 / 3,894.16 / 38,941.63, accuracy 89/90 (98.9%) / evo-t2s Intel CPU (default, loaded window 4096) [projected] vs claude-sonnet-5-5, SWE-Gym median [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over, projected from evo-x2's ollama_ctx_4096 tier]: all-cloud saves 0, accuracy 90/90 (100.0%); naive local saves 79.83 / 798.27 / 7,982.66, accuracy 53/90 (58.9%), 37 wrong answers silent; hybrid flag saves -93.83 / -938.30 / -9,383.02, accuracy 90/90 (100.0%), escalated 88, 0 wrong answers silent; ceiling, no such detector: saves 6.81 / 68.12 / 681.18, accuracy 90/90 (100.0%) / evo-t2s Intel CPU (default, loaded window 4096) [projected] vs claude-sonnet-5-5, B3 v2 tasks (30 tasks x 3 seeds), median [measured; accuracy projected from evo-x2's ollama_ctx_4096 tier]: all-cloud saves 0, accuracy 90/90 (100.0%); naive local saves 30.61 / 306.10 / 3,060.99, accuracy 53/90 (58.9%), 37 wrong answers silent; hybrid flag saves -35.98 / -359.80 / -3,597.97, accuracy 90/90 (100.0%), escalated 88, 0 wrong answers silent; ceiling, no such detector: saves 2.61 / 26.12 / 261.20, accuracy 90/90 (100.0%) / evo-t2s Intel CPU (default, loaded window 4096) [projected] vs claude-sonnet-5-5, nebius-rebench-openhands median [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over, projected from evo-x2's ollama_ctx_4096 tier]: all-cloud saves 0, accuracy 90/90 (100.0%); naive local saves 7,592.12 / 75,921.17 / 759,211.69, accuracy 53/90 (58.9%), 37 wrong answers silent; hybrid flag saves -8,923.97 / -89,239.74 / -892,397.43, accuracy 90/90 (100.0%), escalated 88, 0 wrong answers silent; ceiling, no such detector: saves 647.85 / 6,478.51 / 64,785.13, accuracy 90/90 (100.0%) / evo-t2s Intel CPU (default, loaded window 4096) [projected] vs claude-sonnet-5-5, nebius-rebench-openhands p90 [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over, projected from evo-x2's ollama_ctx_4096 tier]: all-cloud saves 0, accuracy 90/90 (100.0%); naive local saves 13,016.01 / 130,160.11 / 1,301,601.08, accuracy 53/90 (58.9%), 37 wrong answers silent; hybrid flag saves -15,299.36 / -152,993.62 / -1,529,936.21, accuracy 90/90 (100.0%), escalated 88, 0 wrong answers silent; ceiling, no such detector: saves 1,110.68 / 11,106.84 / 111,068.35, accuracy 90/90 (100.0%)

`impact-P3` (VERIFIED, n 3)

claude-sonnet-5-5 standard caching on, sessions shaped like llama3.1:8b's: status measured; measured USD/day 3.60, assumed USD/day 0.00, range USD/day 3.60 to 71.96; per session: re-sent history 1,798,884 tok, USD 0.1799 caching on, 3.5978 no caching; whole session to the cloud USD 0.3358 caching on, 3.7302 no caching (n=5); assumptions: team size = 10 (assumption (profile JSON)); agent sessions per developer per day = 4 (assumption (profile JSON)); cloud session share = 0.5 (assumption (profile JSON)); cloud model = claude-sonnet-5-5 standard, caching on (results/cloud_prices_anthropic_20261014.json (official Anthropic pricing page)); token basis = o200k_base counts (assumption: Anthropic tokenizer counts not measured); re-sent history is a cacheable prefix = R2 sessions are append-only (best case for caching; real histories not measured); confidence high

`impact-hour-value` (VERIFIED, n 1)

USD 64.44/h ($64.44 per hour; Quick Facts: Software Developers, Quality Assurance Analysts, and Testers, row "2025 Median Pay"; May 2025; https://www.bls.gov/ooh/computer-and-information-technology/software-developers.htm; fetched 2026-10-14)

`impact-session-time-by-window` (VERIFIED, n 75)

evo-x2 R2 sessions (40 turns), median wall time per session: llama3.1:8b: ollama_default median 43.3 min/session (n=5), num_ctx_4096 median 7.0 min/session (n=5), ratio 0.16; qwen3:14b: ollama_default median 202.3 min/session (n=5), num_ctx_4096 median 12.2 min/session (n=5), ratio 0.06; qwen3:8b: ollama_default median 71.3 min/session (n=5), num_ctx_4096 median 10.7 min/session (n=5), ratio 0.15

`impact-total` (VERIFIED, n 7)

status quo ollama_default, claude-sonnet-5-5 caching on: measured-only USD/day 3.60 (USD/month 75.55); range USD/day 3.60 to 192.81 (USD/month 75.55 to 4,049.08; low = measured only, nobody waits for compute; high = compute time at the full hour value and cloud without caching) over P2, P3, P4, P5, P6, P7; not added (assumed or not measured): P1; ranking (caching on, time at its upper bound) P5 > P6 > P2 > P3 > P4 > P7 > P1; per row status P1 assumed, P2 mixed, P3 measured, P4 mixed, P5 mixed, P6 mixed, P7 measured

`late-escalation-cost-vs-fresh` (VERIFIED, n 197 redo tasks, 90 fresh tasks, 119 matched rows)

claude-sonnet-5-5, USD per task (cache-priced): late full-history escalation of a wrong local task (cloud-redo-full-context) mean $0.0227, median $0.0169 (n=197) vs a fresh all-cloud run (all-cloud-b3-cost) mean $0.0102, median $0.0090 (n=90); ratio 2.22x on the mean, 1.87x on the median; matched on (seed, task_id) where both exist (seeds 20260901, 20260902, 20260903): 119 redo rows over 53 distinct tasks, redo mean $0.0226, median $0.0169 (n=119) vs fresh of the same task mean $0.0129, median $0.0103 (n=119), ratio 1.76x mean, 1.64x median; incomplete: fixed 167/197 (84.8%), 95% Wilson CI 79.1% to 89.1%; of the 30 unfixed, 28 inherited the local model's earlier executed actions (earlier_state_repair), cloud_model_error 2; evo-t2s Intel CPU (default, loaded window 4096) [projected, from evo-x2's ollama_ctx_4096 tier] vs claude-sonnet-5-5 (hybrid-savings-tradeoff, B3 v2 tasks): the window-flag hybrid escalates 88 of 90 tasks and saves USD -35.98 / -359.80 / -3,597.97/month at 100 / 1,000 / 10,000 tasks/day, accuracy 90/90 (100.0%) vs all-cloud 90/90 (100.0%): it costs more than all-cloud

`num-keep-guardrail-test` (VERIFIED, n 120 turns)

llama3.1:8b num_ctx 4096 num_keep=system: S per seed {20260901: 601, 20260902: 594, 20260903: 597}; observed context shifts 105 with n_keep [595, 598, 602], task n_keep [594, 597, 601], token cuts 7 with keep [594, 601], –keep [], Ollama message-drop calls 224 of 239; rules in use: rule1 94.2% (n=120, first failure 31/-/5); rule3 100.0% (n=120, first failure -/-/-); rule4 100.0% (n=60, first failure -/-/-); canary misses C 6, H 24 of 24 checks; error turns 2; verdict: one-line fix

`num-keep-qwen3-14b` (VERIFIED, n 120 turns)

qwen3:14b num_ctx 4096 num_keep=system: S per seed {20260901: 752, 20260902: 747, 20260903: 750}; observed context shifts 0 with n_keep [], task n_keep [747, 750, 752], token cuts 0 with keep [], –keep [], Ollama message-drop calls 228 of 240; rules in use: rule1 100.0% (n=120, first failure -/-/-); rule2 100.0% (n=120, first failure -/-/-); rule3 95.8% (n=120, first failure -/20/20); rule4 100.0% (n=60, first failure -/-/-); rule5 98.3% (n=120, first failure 5/-/5); canary misses C 1, H 24 of 24 checks; error turns 0; verdict: inconclusive (flag effect not observed)

`t2s-config-comparison` (VERIFIED, n 4 device x model rows)

evo-t2s configuration, measured R2 data (evo-t2s R2 real, 5 seeds x 40 turns); to be replaced by the evo-t2s B5 results (2026-10-26 to 2026-11-08) when they land; kept local, measured by R2 session pass rate (not B3 task accuracy) = share of R2 sessions with no gated silent failure (t2s-r2-real-cpu, t2s-r2-real-igpu); savings = B x k - X with B the measured all-cloud B3 per-task cost and X = (1 - k) x B x (escalation USD per escalated task / all-cloud USD per task - 1): 0 for a tier whose escalated task is costed at its all-cloud price (interim), positive for a tier costed at the measured full-history redo price (see hybrid-savings-tradeoff for which applies) (ASSUMPTION: R2 session share stands for the task share); 30-day month; naive: Intel CPU, Ollama default llama3.1:8b: loaded window 4096, gated silent 5/5 sessions, kept local (measured by R2 session pass rate (not B3 task accuracy)) 0/5 = 0.00; vs claude-haiku-5-5: saves USD 0.00 at 100/day, 0.00 at 1,000/day, 0.00 at 10,000/day/month; vs claude-sonnet-5-5: saves USD -37.49 at 100/day, -374.93 at 1,000/day, -3,749.31 at 10,000/day/month / naive: Intel CPU, Ollama default qwen3:14b: loaded window 4096, gated silent 5/5 sessions, kept local (measured by R2 session pass rate (not B3 task accuracy)) 0/5 = 0.00; vs claude-haiku-5-5: saves USD 0.00 at 100/day, 0.00 at 1,000/day, 0.00 at 10,000/day/month;

vs claude-sonnet-5-5: saves USD -37.49 at 100/day, -374.93 at 1,000/day, -3,749.31 at 10,000/day/month / recommended: Intel iGPU, Ollama default llama3.1:8b: loaded window 32768, gated silent 0/5 sessions, kept local (measured by R2 session pass rate (not B3 task accuracy)) 5/5 = 1.00; vs claude-haiku-5-5: saves USD 2.01 at 100/day, 20.12 at 1,000/day, 201.25 at 10,000/day/month; vs claude-sonnet-5-5: saves USD 30.61 at 100/day, 306.10 at 1,000/day, 3,060.99 at 10,000/day/month / recommended: Intel iGPU, Ollama default qwen3:14b: loaded window 32768, gated silent 1/5 sessions, kept local (measured by R2 session pass rate (not B3 task accuracy)) 4/5 = 0.80; vs claude-haiku-5-5: saves USD 1.61 at 100/day, 16.10 at 1,000/day, 161.00 at 10,000/day/month; vs claude-sonnet-5-5: saves USD 16.99 at 100/day, 169.89 at 1,000/day, 1,698.93 at 10,000/day/month

`t2s-r2-mitigation-cpu` (VERIFIED, n 6 sessions, 240 turns)

overall refuted; num_ctx_4096 llama3.1:8b recovered; num_ctx_4096 qwen3:14b not_recovered (rules below 90%: rule2_log_event_called)

`t2s-r2-real-cpu` (VERIFIED, n 10 sessions)

llama3.1:8b: loaded window [4096], first over window turn [3]; gated 5/5 silent, 5 after window exceeded, 0 baseline-type (gate excludes recall); intact 0/0/0/0/0/0/0/0 / qwen3:14b: loaded window [4096], first over window turn [3]; gated 5/5 silent, 5 after window exceeded, 0 baseline-type (gate excludes recall); intact 0/0/0/0/0/0/0/0; pre-registered kill criterion killed=False, ollama_default 10/10 silent

`t2s-r2-real-igpu` (VERIFIED, n 10 sessions)

llama3.1:8b: loaded window [32768], first over window turn [23]; gated 0/5 silent, 0 after window exceeded, 0 baseline-type (gate excludes nothing); intact 5/5/5/4/4/0/0/0 / qwen3:14b: loaded window [32768], first over window turn [20, 21]; gated 1/5 silent, 0 after window exceeded, 1 baseline-type (gate excludes recall); intact 4/4/4/4/0/0/0/0; pre-registered kill criterion killed=False, ollama_default 5/10 silent

`x2-evalset-tasks-real-v2-ci-units` (VERIFIED, n 25 sessions, 750 task turns)

qwen3:14b wrong-task excess over default; ollama_ctx_16384: excess +1.3 points; session-level 95% CI -1.3 to +4.0 (5 vs 5 sessions); turn-level 95% CI -6.0 to +8.7 (150 vs 150 task turns); ollama_ctx_8192: excess +24.0 points; session-level 95% CI +18.7 to +29.3 (5 vs 5 sessions); turn-level 95% CI +14.7 to +33.3 (150 vs 150 task turns); ollama_ctx_4096: excess +30.7 points; session-level 95% CI +26.0 to +34.7 (5 vs 5 sessions); turn-level 95% CI +21.3 to +40.0 (150 vs 150 task turns); ollama_ctx_4096_numkeep: excess +18.7 points; session-level 95% CI +16.0 to +21.3 (5 vs 5 sessions); turn-level 95% CI +10.0 to +27.3 (150 vs 150 task turns) (10000 resamples each, seed 20261014; the paper uses the session-level CI)

`x2-evalset-tasks-real-v2-mistral-small32-wrong-rate` (VERIFIED, n 0 task turns)

run finished False; mistral-small3.2:24b: refused (mistral-small3.2:24b: negative control failed (canary misses 16 at num_ctx 131072); mistral-small3.2:24b: task tool called correctly on 30/81 baseline turns (< 90%, task-tool gate); mistral-small3.2:24b: task end-state gate failed (33/90 < 90%); mistral-small3.2:24b: strict preflight, gates not passed: ['baseline', 'negative_control', 'task_tool', 'task_end_state']) (bootstrap 10000 resamples, unit session, stratified by tier, seed 20261014)

`x2-evalset-tasks-real-v2-qwen3-8b-wrong-rate` (VERIFIED, n 0 task turns)

run finished False; qwen3:8b: refused (qwen3:8b: task tool called correctly on 61/81 baseline turns (< 90%, task-tool gate); qwen3:8b: task end-state gate failed (65/90 < 90%); qwen3:8b: strict preflight, gates not passed: ['baseline', 'task_tool', 'task_end_state']) (bootstrap 10000 resamples, unit session, stratified by tier, seed 20261014)

`x2-evalset-tasks-real-v2-wrong-rate` (VERIFIED, n 750 task turns)

run finished True; llama3.1:8b: refused (llama3.1:8b: negative control failed (canary misses 26 at num_ctx 131072); llama3.1:8b: not in this plan's model list ['qwen3:14b']; llama3.1:8b: task end-state gate failed (72/90 < 90%)) / qwen3:14b: valid True; ollama_default: wrong 17/150 (11.3%) (silent 17, surfaced 0), loaded context [40960]; ollama_ctx_16384: wrong 19/150 (12.7%) (silent 19, surfaced 0), loaded context [16384]; ollama_ctx_8192: wrong 53/150 (35.3%) (silent 53, surfaced 0), loaded context [8192]; ollama_ctx_4096: wrong 63/150 (42.0%) (silent 63, surfaced 0), loaded context [4096]; ollama_ctx_4096_numkeep: wrong 45/150 (30.0%) (silent 45, surfaced 0), loaded context [4096], num_keep per seed {20260901: 3369, 20260902: 3369, 20260903: 3364, 20260904: 3369, 20260905: 3367}; ollama_ctx_16384 excess over default +1.3 points (95% CI -1.3 to +4.0, 5 vs 5 sessions); reading: no excess shown; ollama_ctx_8192 excess over default +24.0 points (95% CI +18.7 to +29.3, 5 vs 5 sessions); reading:

ollama_ctx_8192 raises the wrong-task rate; ollama_ctx_4096 excess over default +30.7 points (95% CI +26.0 to +34.7, 5 vs 5 sessions); reading: ollama_ctx_4096 raises the wrong-task rate; ollama_ctx_4096_numkeep excess over default +18.7 points (95% CI +16.0 to +21.3, 5 vs 5 sessions); reading: ollama_ctx_4096_numkeep raises the wrong-task rate (bootstrap 10000 resamples, unit session, stratified by tier, seed 20261014)

`x2-scale-real-default-tier` (VERIFIED, n 3 sessions)

llama3.3:70b: no default-tier session / qwen3:32b: 3 default-tier sessions completed (seed 20260901: 40 turns, 6.18 h, loaded [40960], first over window 24; seed 20260902: 40 turns, 6.01 h, loaded [40960], first over window 25; seed 20260903: 40 turns, 4.66 h, loaded [40960], first over window 26)

`x2-v3-usability` (VERIFIED, n 121)

per-call timeout 900 s (harness DEFAULT_CALL_TIMEOUT_S); timeout_latency/done (share) [context tiers]: llama3.1:8b/llama_server 40/340 (11.8%) [longdoc_120000 x10, longdoc_64000 x10, longdoc_80000 x10, longdoc_96000 x10]; llama3.1:8b/ollama_default 10/340 (2.9%) [longdoc_120000 x10]; qwen3-14b/llama_server 0/340 (0.0%); qwen3-14b/ollama_default 0/340 (0.0%); qwen3-30b-a3b/llama_server 10/340 (2.9%) [longdoc_120000 x10]; qwen3-30b-a3b/ollama_default 0/340 (0.0%); qwen3-32b/llama_server 11/100 (11.0%) [longdoc_32000 x10, trace_mix_24000 x1]; qwen3-32b/ollama_default 0/100 (0.0%); qwen3-4b-2507/llama_server 40/340 (11.8%) [longdoc_120000 x10, longdoc_64000 x10, longdoc_80000 x10, longdoc_96000 x10]; qwen3-4b-2507/ollama_default 10/340 (2.9%) [longdoc_120000 x10]; qwen3-8b/llama_server 0/340 (0.0%); qwen3-8b/ollama_default 0/340 (0.0%)